

Comparison of the PHISICS/RELAP5-3D Ring and Block Model Results for Phase I of the OECD/NEA MHTGR-350 Benchmark

G. Strydom, A. S. Epiney,
A. Alfonsi, C. Rabiti

January 2016

The INL is a
U.S. Department of Energy
National Laboratory
operated by
Battelle Energy Alliance



This is an accepted manuscript of a paper intended for publication in a journal. This document was prepared as an account of work sponsored by an agency of the United States Government. Neither the United States Government nor any agency thereof, or any of their employees, makes any warranty, expressed or implied, or assumes any legal liability or responsibility for any third party's use, or the results of such use, of any information, apparatus, product or process disclosed in this report, or represents that its use by such third party would not infringe privately owned rights. The views expressed in this paper are not necessarily those of the United States Government or the sponsoring agency.

Prepared for the U.S. Department of Energy
Office of Nuclear Energy
Under DOE Idaho Operations Office
Contract DE-AC07-05ID14517

Comparison of the PHISICS/RELAP5-3D Ring and Block Model Results for Phase I of the OECD/NEA MHTGR-350 Benchmark

G. STRYDOM ^{a*}, A. S. EPINEY ^b, A. ALFONSI ^a, C. RABITI ^a

^a *Idaho National Laboratory: Nuclear Science and Technology Division*

2525 N. Fremont Avenue, Idaho Falls, Idaho 83415

Tel: (208) 526-1216, Fax: (208) 526-2930, Email:gerhard.strydom@inl.gov

^b *Paul Scherer Institute: Laboratory for Reactor Physics and Systems Behaviour (LRS)*

5232 Villigen-PSI, Switzerland

Number of pages: 58 total, 28 text only

Number of tables: 6

Number of figures: 23

* Corresponding author

Keywords:

RELAP5-3D, MHTGR-350, OECD, PHYSICS

Abstract

The PHISICS code system has been under development at INL since 2010. It consists of several modules providing improved coupled core simulation capability: INSTANT (3D nodal transport core calculations), MRTAU (depletion and decay heat generation) and modules performing criticality searches, fuel shuffling and generalized perturbation. Coupling of the PHISICS code suite to the thermal hydraulics system code RELAP5-3D was finalized in 2013, and as part of the verification and validation effort the first phase of the OECD/NEA MHTGR-350 Benchmark has now been completed.

The theoretical basis and latest development status of the coupled PHISICS/RELAP5-3D tool are described in more detail in a concurrent paper. This paper provides an overview of the OECD/NEA MHTGR-350 Benchmark and presents the results of Exercises 2 and 3 defined for Phase I. Exercise 2 required the modelling of a stand-alone thermal fluids solution at End of Equilibrium Cycle for the Modular High Temperature Reactor (MHTGR). The RELAP5-3D results of four sub-cases are discussed, consisting of various combinations of coolant bypass flows and material thermophysical properties. Exercise 3 required a coupled neutronics and thermal fluids solution, and the PHISICS/RELAP5-3D code suite was used to calculate the results of two sub-cases.

The main focus of the paper is a comparison of results obtained with the traditional RELAP5-3D “ring” model approach against a much more detailed model that include kinetics feedback on individual block level and thermal feedbacks on a triangular sub-mesh. The higher fidelity that can be obtained by this “block” model is illustrated with comparison results on the temperature, power density and flux distributions. It is shown

that the ring model leads to significantly lower fuel temperatures (up to 10%) when compared with the higher fidelity block model, and that the additional model development and run-time efforts are worth the gains obtained in the improved spatial temperature and flux distributions.

Keywords: HTGR, RELAP5-3D, PHISICS

I. INTRODUCTION

The High Temperature Gas-Cooled Reactor (HTGR) Methods Core Simulation group at the Idaho National Laboratory (INL) leads the development of the Organization for Economic Cooperation and Development (OECD) MHTGR-350 Transient Benchmark for evaluating prismatic HTGR analysis codes. The benchmark is sponsored by the OECD's Nuclear Energy Agency, and will eventually yield a set of peer-reviewed steady-state, transient, and lattice solutions that can be used by the Department of Energy, Nuclear Regulatory Commission and vendors to assess their codes. The technical specification is based on the General Atomics 350 MW Modular High Temperature Gas-Cooled Reactor (MHTGR) design ^{1,2}. The HTGR Methods group at INL is responsible for defining the benchmark specifications, lead the data collection and comparison activities, and chair the annual technical workshops. Several code development and verification activities from participants in Germany, Korea and the United States are currently based on the MHTGR-350 benchmark, and progress on the INL contributions have already been completed ^{3,4,5}.

As part of the benchmark simulation activities, the Parallel and Highly Innovative Simulation for INL Code System (PHISICS) ⁶ reactor physics package was coupled to the INL-developed system thermal analyses code RELAP5-3D ⁷ to enable neutron transport calculations for the core as well as more flexible cross section handling in

coupled core simulations. RELAP5-3D typically does not provide the spatial resolution of higher fidelity thermal fluid codes but it does have a well-validated plant simulation capability.

This paper summarizes the INL results obtained for Exercises 2 and 3 defined for Phase I of the benchmark, with a specific focus on a comparison of the traditional homogenized (ring) model approach with a much more detailed triangular geometry model for the thermal-hydraulic solution with RELAP5-3D. These two models were developed during the coupling of the PHISICS/RELAP5-3D system to investigate the effect of spatial refinements in the temperature fields on the typical parameters of interest (fuel temperatures, k_{eff}), and to assess the suitability of utilizing a systems code like RELAP5-3D to provide detailed temperature profiles.

II. OECD/NEA MHTGR-350 BENCHMARK OVERVIEW

The MHTGR-350 MW benchmark specification was developed in cooperation with General Atomics to compare different codes and methods, since validation data from current or past prismatic HTGR experimental facilities or reactors are severely limited. The scope of the benchmark is to establish a well-defined problem based on a common given data set, and to compare methods and tools in core simulation and thermal hydraulic analysis through a set of multi-dimensional computational test problems. The MHTGR-350 MW benchmark is based on the approach followed for the OECD/NEA PBMR-400 MW benchmark⁸. It consists of three Phase I steady-state exercises, four Phase II transient exercises, and two Phase III depletion exercises. The focus of this RELAP5-3D comparison paper is on the results obtained for the Exercises 2 and 3 of

Phase I, as summarized below. (The stand-alone neutronics results obtained for Exercise 1 were included in a 2013 INL status report ⁴, and will not be discussed here).

- **Exercise 1:** Neutronics only steady-state solution for the 350 MW End of Equilibrium Cycle (EOEC) MHTGR core, using the specified geometry, material descriptions, and detailed cross-section libraries.
- **Exercise 2:** Thermal hydraulics only steady-state core solution. Four sub-cases are defined, depending on the core bypass flow type and the use of fixed or variable thermophysical material properties. Participants are expected to provide steady-state solutions for each of these sub-cases according to their codes' capabilities. A defined core power distribution map must be used, since no neutronics are involved in this exercise.
- **Exercise 3:** Coupled neutronics-thermal hydraulic core steady state. This exercise is a combination of the first two exercises and the coupled steady-state solution must be calculated using the provided temperature dependent cross-section library and burnup distributions.

The axial and radial core layouts of the MHTGR-350 design are shown in Fig. 1 and Fig. 2, respectively, and the major core characteristics are summarized in Table I. (In Fig. 1, the Control Rods (CR), Core Barrel (CB) and Reactor Pressure Vessel (RPV) are also indicated). A detailed description of the MHTGR-350 design can be found in the benchmark specification document (currently still in draft format ²). The 1/3rd core numbering utilized for the benchmark data reporting, and the discussion that follows in this paper, is shown in Fig. 3 for the first (bottom) layer of the core and reflector regions.

III. CODE AND MODEL DESCRIPTION

This section provides an overview of the simulation codes utilized for this benchmark, as well as a discussion of the models developed and major assumptions that were made.

III.A. The Physics Code Suite

The complex phenomena in Generation IV systems such as the MHTGR design challenge legacy system codes like RELAP5-3D, which were developed for best estimate transient simulation of light water reactors ⁷. RELAP5-3D is able to model coupled behaviour of the plant system (heat exchangers, steam generators, pumps, valves, etc.), thermal-hydraulics of the reactor core and the neutronics of the reactor. The code was specifically designed for simulations of light water reactor transients such as loss of coolant, anticipated transients without scram, and operational transients such as loss of feed water, etc. The multidimensional neutron kinetics model in RELAP5-3D is based on the NESTLE code ⁹, which solves the two or four group neutron diffusion equations in either Cartesian or hexagonal geometry. A diffusion solution of the flux in four energy groups may however not be sufficient to get an accurate representation of the power distribution for the MHTGR ¹⁰. In addition, linear cross section feedbacks on thermal hydraulic state parameters (such as fuel and moderator temperatures) as supported by NESTLE/RELAP5-3D are not detailed enough to capture the complex MTHGR cross section behaviour.

To be able to simulate MHTGR transients with the required neutronic fidelity, the neutronic code package PHISICS has been coupled to RELAP5-3D. PHISICS (Parallel and Highly Innovative Simulation for INL Code System) is a neutronics code system in development ⁶ at INL. The different modules for PHISICS are a nodal and semi-

structured spherical harmonics based transport core solver (INSTANT ¹¹) for steady state and time dependent problems, a depletion module (MRTAU ¹²) and a cross section mixer-interpolator (MIXER ¹³) module. A detailed discussion of PHISICS' theory and structure are provided for interested readers¹⁴. This flexible package provides the functionality to cope with the challenges of coupled neutronic/thermal hydraulic MHTGR core simulations.

Each different module of PHISICS contains a kernel (module) that solves a basic problem. A local driver is assigned to each kernel which is able to run it in standalone mode. Communication between the kernels is managed by the use of global data types that hold global information (cross section data, mesh, fluxes, etc.) that are needed by more than one kernel to perform complex calculations involving different kernels. Global drivers solving a complex problem calling different kernels can be developed easily with this flexible software structure. The PHISICS code is still in development to extend its capabilities (see ¹⁴ for more detail).

The transport core solver INSTANT (Intelligent Nodal and Semi-structured Treatment for Advanced Neutron Transport) is the key kernel of the PHISICS framework. INSTANT is parallelized and is designed to take full advantage of medium to large clusters (10 to 1000 processors). It is based on the second order formulation of the transport equation discretized in angle by spherical harmonics while in space it uses orthonormal polynomials of an arbitrary order. In addition to steady-state solutions, INSTANT is able to solve time dependent problems. For that, a scheme based on a second order backward Euler scheme with explicit delayed neutron treatment has been implemented as a new module for the PHISICS suite ⁵.

MRTAU (Multi-Reactor Transmutation Analysis Utility) is a generic depletion code developed at INL ¹². In addition to core depletion, the code can be utilised for stand-alone decay heat calculations. It tracks the time evolution of the isotopic concentration of a given material accounting for nuclear reactions occurring in the presence of neutron flux, and also due to natural decay (Bateman equation). The code uses a Taylor series expansion based algorithm at arbitrary order and the Chebyshev Rational Approximation Method (CRAM) for computation of the exponential matrix.

The MIXER module does all the cross section handling for the different kernels. MIXER can handle macroscopic, microscopic and “mixed” cross sections. A macroscopic cross section library contains macroscopic cross sections for each type of material used in the calculation (fuel, reflector, etc.) tabulated for the state parameters (temperature, burn-up, control rod position, etc.). MIXER interpolates these cross sections at the requested state parameters, and no limits in tabulation dimensions or neutron energy groups exist. A microscopic or “mixed” cross section library contains the tabulated cross sections for each isotope considered in the calculation. MIXER reads a description containing a list of isotopes and corresponding densities for each material, and interpolate the microscopic cross sections at the requested state parameters. Macroscopic cross sections for each material are generated with the corresponding number densities.

With this capability, “mixed” macroscopic and microscopic cross sections are also possible. For example, it is possible to provide macroscopic absorption cross sections without xenon for a material and the microscopic xenon absorption cross section together with a xenon density. MIXER will calculate the xenon absorption contribution and add it to the macroscopic cross section. MIXER can read different cross section library formats. Among them is an original simple XML based format, but also AMPX,

ISOTXS and ECCO library formats that allow cross section libraries to be prepared with SCALE, ERANOS or MC2. More library types are planned to be supported in the future.

III.B. The coupling of PHISICS to RELAP5-3D

In order to increase the neutronic capabilities of RELAP5-3D, a direct coupling with the PHISICS code has been performed, i.e., PHISICS is integrated with RELAP5-3D as a set of subroutines¹⁴. This gives the user access to the full capability of PHISICS from within RELAP5-3D. The PHISICS part of coupled calculations can be parallelized on multiple processors, where the calculation is always driven by RELAP5-3D. The RELAP5-3D input reader decides if parts of PHISICS are needed and calls the PHISICS/RELAP5-3D driver accordingly. The driver collects the available input data from RELAP5-3D (geometry, calculation options) and adds, if needed, data from special PHISICS input files (cross sections, transport solver and depletion parameters, etc.) to complete the required information for the specified calculation. Once all the inputs are collected, the PHISICS/RELAP5-3D driver calls the required modules and feeds back the results (power distribution) to RELAP5-3D for the next iteration. Fig. 4 shows the general input data flows between the two packages.

RELAP5-3D is interfaced with the PHISICS code through a direct software “pipeline” (interface) that manages the exchange of information between the two “physics” elements (TH feedbacks from RELAP5-3D to PHISICS, fission and decay heat power distributions from PHISICS to RELAP5-3D, etc.).

This “direct” coupling allows the user to run existing RELAP5-3D input decks with INSTANT as the core solver (this option is made accessible from within RELAP5-3D

through a new keyword). The coupling is compatible with the existing RELAP5-3D cross section and feedback options. Existing cross section inputs and mappings from kinetic nodes to thermal hydraulic zones can be used with the INSTANT option. The RELAP5-3D control rod model can also be used. In addition, the coupling supports a new cross section option which relaxes the RELAP5-3D limitation of four energy groups (only if INSTANT is used). The number of neutron energy groups is only limited by the available computer memory. Multi-group cross sections for this new option support an unlimited number of tabulation dimensions (fuel temperature, moderator temperature, control rod position, xenon density, etc.) and an unlimited number of tabulation points per dimension. This new cross section option is compatible with existing mappings from kinetic nodes to thermal hydraulic zones in RELAP5-3D.

The depletion code MRTAU is also coupled to RELAP5-3D as part of the PHISICS suite. It can be used to perform core depletion calculations with coupled thermal hydraulic feedbacks. In addition, MRTAU is able to burn the core to a desired depletion level, or to perform cooling (decay), before a transient is initiated. MRTAU can also be used for decay heat calculation and poison tracking during transients. The possible calculation paths resulting by combining INSTANT, MRTAU and RELAP5-3D are manifold ¹⁴. In this paper, a summary of the calculation path used for the OECD/NEA MHTGR-350 MW benchmark are briefly presented. The calculation path can be split into two main classes, namely the “depletion time evolution” and the “time dependent” calculation path.

The “depletion time evolution” scheme is used to compute core depletion with an updated temperature field in the core. Fig. 5a shows the calculation path for this scheme. The first loop (on the left in the figure) is iterated between MIXER, INSTANT and RELAP5-3D until temperature convergence is reached taking into account the

neutronic feedback. The calculation then goes into the depletion loop (shown on the right in the figure). Here, the thermal hydraulic state parameters are kept constant, but the isotopic densities from the depletion calculation are updated. The scheme continues to deplete the core with user-defined time steps until an update of the temperature field is requested by the user (in Fig. 5a, $t=T_n$). This “depletion time evolution” scheme is used in the MHTGR benchmark analysis to find the steady state temperature distribution in the core with xenon at its equilibrium value (see Phase I Exercise 3, Section V).

The “time dependent” calculation scheme with MRTAU is used to track isotopic densities during a coupled thermal hydraulic transient (i.e. RELAP-3D in transient mode). Fig. 5b shows the calculation scheme for the “time dependent” mode. In this mode, MRTAU depletes the core for one time step, as specified by RELAP5-3D. The updated sources and cross sections are then sent to INSTANT which computes the new core power and flux distribution for the current time step. During the next time step RELAP5-3D calculates the new temperature distribution. The MIXER module then finally updates the cross sections for the new thermal hydraulic state variables and sends it to MRTAU for the next depletion step. This scheme is not demonstrated in this paper, but will be utilized for the calculation of time-dependent xenon build-up and decay during the control rod movements defined for Phase II Exercise 4 ².

III.C. Ring and block Model Descriptions

Two different PHISICS/RELAP5-3D models have been set up for the benchmark to investigate the improvements that can be obtained with a refined spatial representation. One models the core as 9 homogenized rings whereas the higher fidelity model is refined to one-sixth of a fuel block (hereafter referred to as the “ring” and “block”

models). The details and assumptions for both these models are briefly presented here, in addition to the overviews presented previously^{3,4,5}. Since the two models only differ in the way they model the core and reflector regions, the reactor vessel nodalization for RELAP5-3D is the same for both models and is shown in Fig. 6. The time-dependent junction #255 adjusts the inflow helium mass flow rate to obtain a target outlet helium temperature of 687°C through a control variable that combines the outlet pressure (6.39 MPa), inlet helium temperature (259°C) and the total thermal power (350 MW) boundary conditions. In addition to the core region, the model includes the lower plenum (#175), coolant riser (#115), upper plenum (#125), outlet plenum (#170), vessel gap and the vessel metal structures. The benchmark does not require the modelling of the secondary system.

In both the ring and block models, PHISICS uses a hexagonal mesh for the neutronics solution corresponding to a fuel or reflector block. One third of the core is modelled to make use of the 120° azimuthal symmetry (see Fig. 1). The core is divided into 14 axial levels, two each for the upper and lower reflector and 10 levels for the core region. The 26 energy group cross-section library is read as separate data structures, and updates are performed on these base cross-section sets during the steady state convergence process in the MIXER module. The cross section updates are performed for four state parameters: moderator and fuel temperature, xenon-135 concentration and the hydrogen concentration (for the water ingress transient).

The four-dimensional cross-section tables are designed to cover the initial steady-state conditions and the expected ranges of the four selected instantaneous feedback parameters in the transients to be simulated in the benchmark. The reference tabulated cross-section set was generated from the DRAGON 4 lattice code using the end of equilibrium core number densities with 26 energy groups².

III.C.1 Ring model

The ring model follows the common system-code homogenization approach of modelling the inner reflector, fuelled core region, and outer reflector as rings in cylindrical coordinates, with three additional rings representing the core barrel (CB), reactor pressure vessel (RPV), and outer air boundary layer (Fig. 7). The coarse axial mesh is discretized in 2 upper reflector, 10 core and 2 lower reflector elements, as shown Fig. 2. The core consists of 9 parallel flow channels representing each a ring in the core, i.e., there are three channels representing the inner reflector, three channels representing the three fuel rings, two channels representing the replaceable outer reflector and one channel representing the permanent outer reflector. Each channel is coupled to its own heat structure representing the fuel or graphite blocks in the corresponding core ring. The radii of the rings are calculated to represent the correct graphite volumes in each of the structures, and it includes corrections for the 2 mm gaps between the reflector blocks and the 3 mm gap between the outer reflector and the metallic core barrel.

In addition, a total of seven core bypass flow channels represent the flow through the gaps between the graphite blocks, according to the specifications of the benchmark.

A number of conduction and radiation sets are included in the model to account for radial conduction and axial radiation between the graphite structures in the inner, outer, top and bottom reflectors. Radial radiation heat transfer is also modelled between the outer reflector surface and the core barrel, and from the outer surface of the reactor vessel to the boundary air layer, as shown in Fig. 8. Adiabatic boundary conditions are applied at the top and bottom model boundaries, and the outer radial air layer is defined to be at a constant temperature of 30°C.

III.C.2 The unit cell

In the MHTGR core design, neutronic feedbacks on the fuel and moderator (graphite) temperatures are quite different. Therefore, it is desirable to have an accurate prediction of the fuel and moderator temperatures independently. To model the fuel region of the core, a cylindrical “unit cell” has been defined, consisting of a fuel compact, its surrounding matrix graphite and a helium coolant channel (Fig. 9). In this approach, the fuel compact (TRISO fuel particles embedded in graphite) is homogenization to a single region. The unit cell is constructed as follows:

- The radius of the coolant hole corresponds to the actual design radius.
- The distance between the coolant hole and the homogenized fuel region, i.e. the thickness of the inner graphite region, corresponds to the actual design distance so that the heat transfer distance is conserved.
- The homogenized fuel region conserves the actual design fuel *volume*, i.e. the thickness of this region is determined by the fuel volume in the unit cell.
- The size of the outer graphite region (outer ring in Fig. 9) is such that the total graphite *mass* in the unit cell is represented.

At this stage detailed results for a representative TRISO fuel particle within the fuel compact is not available, but preliminary investigations have been performed using an additional heat structure in RELAP5-3D to represent the layers within the TRISO kernels. Despite the fact that the fuel compact is still homogenised, the use of a “unit cell” model leads to better fuel and moderator temperatures prediction than a homogenized core ring. In the ring model three unit cells are therefore used (one per

fuel ring) to distinguish between the average fuel and moderator temperature in the inner, center and outer fuel rings.

III.C.3 The block model

In the block model, a single hexagonal block is explicitly represented by dividing it into six triangular elements, as shown in Fig. 10. Each of these triangles is represented by the unit cell heat structure in RELAP5-3D as described above. Each unit cell heat structure is connected to two hydraulic channels; one in the center representing the core coolant helium flow and one on the outer surface representing the gap flow between two fuel blocks. Heat transfer between the triangular elements occurs in two ways: inside a block via conduction, and between blocks via radiation across the 2 mm gap between the block faces. These connections are defined as part of the conduction and radiation enclosure sets in the RELAP5-3D model, in addition to the existing enclosure sets already defined for the ring model (Fig. 8). This block model can therefore provide a much more detailed temperature distribution compared to the ring model.

In the case of the first fuel ring, for example, blocks 8-13 in Fig. 3 can now be represented with $6 \times 6 = 36$ triangular elements, which provide 36 data points per axial layer compared to only one value for the ring model.

In theory, one could imagine representing the whole core including the reflector regions ($91 \times 14 = 1274$ blocks) using the triangular block representation. Unfortunately, RELAP5-3D has some restrictions that lead to a limit on the fidelity that can be obtained:

- Neutronic feedback zones:** A feedback zone defines how feedback parameters (temperatures, densities, etc.) calculated by RELAP5-3D are averaged before passed to the neutronic solver as well as how the power computed in a node by the neutronic solver is distributed among the RELAP5-3D heat structures. The modelling of Ex. 3 required the use of kinetic feedback zones to manage the exchange of power (calculated by INSTANT) and temperature (RELAP5-3D) data. There is however a limit on the number of these data types that the code can process. It was initially envisaged to couple the triangular feedback elements on a one-to-one basis, i.e., one triangular node in RELAP5-3D would provide temperature data to a triangular node in INSTANT, which would then pass back the triangular power distribution to RELAP5-3D for the next iteration. Since the modelling of the full MHTGR-350 core graphite structures in a triangular geometry requires 7,644 kinetic nodes, a full core triangular model was not an option due the limitations in the number of kinetic zones available in RELAP5-3D. (A total of $7,644 \times 2 = 15,288$ nodes are actually required if the both the fuel and moderator temperature are to be resolved separately). On the neutronics side of the coupling, INSTANT therefore uses a hexagonal mesh for the neutronics solution, i.e. corresponding to a single fuel or reflector block.
- Heat conduction/radiation enclosures:** A heat conduction/radiation enclosure in RELAP5-3D contains all surfaces that conduct or radiate heat to each other. Due to input limitations, an enclosure in RELAP5-3D can currently only hold 99 surfaces. To put all surfaces in the core into one enclosure is therefore not possible. Furthermore, there can be only 99 enclosures in total in the model and finally, a heat structure

surface can only be part of one conduction or one radiation enclosure, but not both.

These limitations lead to the following model choices:

- It is not possible to represent the whole core with one-sixth blocks. It was therefore decided to focus the RELAP5-3D triangular sub-division on the important regions of the active core region (Rings 4-6) and the two reflector rings next to the core (Rings 3 and 7). The RELAP5-3D block model (Fig. 11) therefore consists of “smeared-out” representation in Rings 1 and 2 (Blocks 1-3) and 8-10 (Blocks 44-91), and a detailed triangular resolution in Rings 3-7 (Blocks 8-43) (the block numbering is indicated in Fig. 3) .
- It’s not possible to connect one heat structure surface to a conduction and radiation enclosure at the same time. Therefore, it has been decided to use the left side (inner) of each unit cell to model the conduction inside a block and the right side (outer) to model the radiation between blocks. This approach leads to two important unit cell model assumptions. First, it is assumed that the outer side temperature of the unit cell represents the temperature at the outer surface of a block. This temperature is used to radiate the heat between blocks. Second, the inner surface temperature of the unit cell has to be used to compute the conduction between two one-sixth elements of a block. To minimize the impact of these (obviously approximate) assumptions, the model view factors (radiation) and conduction lengths (conduction) have been corrected accordingly.

III.C.4 Model Size and Computational Performance

As can be expected, the increase in model resolution and subsequent larger temperature and power datasets resulted in significant penalties in terms of model size and computational performance. A comparison of the two PHISICS/RELAP5-3D models is presented in Table II. The RELAP5-3D run time of a typical converged steady-state solution for the thermal fluids standalone Ex. 2 increased from 0.1 h (ring model) to 5 h (block model). For Ex. 3, the coupled PHISICS/RELAP5-3D solution required 11 h to converge for the ring model, and 20 h for the block model. The run time for Ex. 3 includes the use of 28 parallel processors, compared to a single processor for Ex. 2. The main question is therefore if this investment in model development, simulation time and post-processing lead to significant gains in the block model results, compared to the much simpler and faster ring model.

IV. BENCHMARK RESULTS: PHASE I EXERCISE 2

RELAP5-3D in stand-alone mode has been used to solve the four sub-cases of Exercise 2, which are defined as thermal fluids standalone calculations with fixed power density profiles in the fuel region. The four sub-cases are defined in order of increasing complexity (and summarized in Table III). Ex. 2a assumes that all coolant flow is through the core region, and only fixed thermophysical properties are specified, as shown in Table IV. For Ex. 2b, the bypass flow modeling in seven fixed bypass flow channels, totalling up to 11% of the total inlet flow rate (Table V), are required. Instead of the fixed thermophysical properties provided for Ex. 2a and Ex. 2b, the material properties prescribed for Ex. 2c vary as complex functions of fluence and temperature². The definition of Ex. 2d combines the variable material properties with the requirement

to model the bypass flow distribution explicitly through the 2-mm gaps between the blocks.

The primary parameters of interest for Ex. 2 are the gas and solid temperatures and global parameters such as the pressure drop over the core and the inflow mass flow rate. The benchmark does however require participants to also report secondary parameters such as the calculated thermal conductivities, mass flow rate or velocities in bypass channels and heat transfer factors, since this data can be utilized to trace the source of possible differences in the primary parameters. In this paper, only a subset of the available data will be presented to illustrate the main characteristics for each of the exercises. Since the variances between Ex. 2a and Ex.2b - Ex.2d is relatively small, most of the data discussed in this section is for Ex. 2a.

IV.A Exercise 2a

The Ex. 2a fuel temperature distributions shown in Fig. 12 (ring model) and Fig. 13 (block model) for the bottom core axial layer illustrate the increased spatial resolution of the block model over the ring model. A more detailed distribution of temperatures was obtained on the triangular mesh compared to three values per axial layer for the ring model. The ring model predicted significantly lower temperatures compared to the block model, which show for example radial temperature variations up to 300°C between the three fuel rings on this axial level (Fig. 13). This is because the block model is not a spatial refinement of the ring model (in the CFD-sense), but an independent model developed with its own assumptions and approximations. The treatment of radiation enclosures, conduction between the triangular elements and conduction individual gas flow connections to each face of the triangles in the block model are significantly different from the much simpler representation in the ring model, where one unit cell

represents an entire axial layer of blocks. For example, since the power density is specified per block, the spatial power variation between two blocks in the same ring can be distinguished in the block model, leading to individual predictions of the temperatures in these blocks. In contrast, the power density in the ring model are volume-averaged over all 6 blocks in the inner ring (for example), which directly leads to an averaging effect on the fuel temperatures as well. It can therefore not necessarily be deduced that this model provides the most accurate temperature predictions without comparison to experimental values.

The two examples presented here show how the local temperature hot spots (e.g. in the inner fuel ring fresh fuel blocks) and variances between the blocks can be identified using a higher resolution model. This capability is lacking in the ring model, and the resultant under-prediction can be important in design and safety studies, where the margins to fuel temperature limits need to be quantified. A more detailed spatial quantification of the peak fuel temperatures during loss of cooling or control rod withdrawal transients could assist in calculating the volume of fuel exposed to high temperatures and power densities, which are typically used as boundary conditions by fuel performance codes for fission product release calculations. This in turn might allow a core designer or utility to license a design for higher power levels, if it can be shown that the margins are sufficient. The additional effort required for the development of a more detailed modelling approach can therefore be justified based on the intended use and possible gains that can be derived from these models.

The graphite temperature distribution is shown for the whole core region in Fig. 14 (ring model) and Fig. 15 (block model). The relatively high inner reflector temperatures can be seen in both figures, since Ex. 2a did not include any bypass flow modelling in the reflectors. The block model (Fig. 15) show six equal values in the regions where a

ring representation is used, e.g. blocks 1 and 2, but the detailed information available from this model is again apparent in the triangular regions of the core.

As an example of the data that can be obtained from the detailed block model, the average fuel temperature axial profiles for the six triangular elements of block 9 are presented in Fig. 16. The block 9 average value can then be calculated from this data, and compared with the inner fuel ring temperature result obtained from the ring model, as shown in Fig. 16. The lower temperatures of the ring model (up to 10% in certain regions) are again apparent in this axial temperature profile, caused by the lack of individual block power density resolution.

IV.B Exercises 2b-2d

The inclusion of the 11% bypass flow in the Ex. 2b RELAP5-3D model leads to significant decreases in the inner and outer graphite reflector temperatures, as shown in Fig. 17. The block model predicts up to 41% lower temperatures in the inner reflector region. The effect on the fuel region is in the opposite direction (i.e., Ex. 2b exhibits higher fuel temperatures), but this difference is comparatively small (less than 10%). This is because a decrease from 100% to 89% of the nominal helium flow rate through the core still represents a very effective convective heat removal mechanism. It should be noted that Ex. 2b represents an unphysical MHTGR-350 scenario, since approximately 6% of all energy is deposited outside the fuel region in the reflectors via fast neutron and gamma interactions ¹⁵. This non-local heat deposition is not included in the benchmark specification, and will lead to much higher reflector temperatures, which will counter the effects of these reflector bypass flows to some degree.

The only difference between Ex. 2b and Ex. 2c is the addition of the variable thermophysical properties. The RELAP5-3D MHTGR-350 benchmark model for Ex. 2c uses dependent thermophysical properties for graphite to account for the fluence, as prescribed in the benchmark specification. The proper modeling of the decrease in thermal conductivity with an increase in fast flux exposure is especially important for fuel and graphite temperatures during the loss of cooling transients in Phase II. Only minor differences are observed between these two Phase I cases, since the convective heat transport of the active helium cooling during normal operation dominates the effects of changes in the material conductivity and specific heat. The difference between Ex. 2b and Ex. 2c will become more significant during the loss of cooling transients that will be performed for Phase II. The results are therefore not shown here.

Exercise 2d combines the effects of bypass flows and variable thermophysical properties. The difference between Ex. 2c and Ex. 2d is the specification that the real geometry of the MHTGR-350 must be used to determine the converged bypass flows in the 2 mm gaps that exist between all blocks, as well as the 3.5 mm gap between the outer reflector and the CB. This exercise is designed to capture the high resolution and fidelity that Computational Fluid Dynamics (CFD) is capable of, but since the intended design use of RELAP5-3D is typically thermal system analysis of a lower fidelity, the current RELAP5-3D model can only approximate the real geometry of these gaps with a network of 1D pipe elements. The RELAP5-3D block model predicts a total of 14.2% bypass flow outside the core region, i.e. relatively close to the 11% value used in Ex. 2b and Ex. 2c.

V. BENCHMARK RESULTS: PHASE I EXERCISE 3

Exercise 3 combines the solutions of the neutronics and thermal fluid domains, and therefore requires the use of a coupled solution methodology. The Ex. 3a model is created by combining the RELAP5-3D thermal fluid model developed for Ex. 2c (11% bypass flow, variable thermophysical properties) with the PHISICS kinetics, while Ex. 2d is used as the basis for Ex. 3b. In the PHISICS/RELAP5-3D solver scheme, the xenon equilibrium is obtained by using MRTAU to “burn” the core in 4 time steps of 2.5 days each, i.e. for a total of 10 days. This process involves four recalculations of the flux with INSTANT after every 2.5 days, using the scheme illustrated in Fig. 5a. A comparison between the eigenvalues and Control Rod (CR) worths obtained for Ex. 3a and Ex. 3b is summarized in Table VI for the ring and block models.

For Ex. 3a, the difference between the ring and block models are relatively small (less than 160 pcm), and the predicted CR worths are within 80 pcm for the two models. Since the neutronics model is the same for the ring and block cases, these variances are caused by the refined feedbacks provided by RELAP5-3D. The block model eigenvalues are less than the ring model values for Ex. 3b, and the two CR worths are almost identical. The only differences between the two cases (3a and 3b) are the changes in the fuel (Doppler) and moderator temperatures due to the slight variation in the bypass flow distribution, which act as the drivers for the small variances predicted for the two exercises.

The power density distribution reaches a maximum axial value on level 8 (634 cm from the bottom of the core). The power density distribution on level 8 is shown for Ex.

3a in Fig. 18, where a peak value of 11 W/cm^3 is calculated in block 13. (The power and flux data are calculated on the hexagonal mesh, as explained in section III.C).

A fast and a thermal flux data example is shown in Fig. 19 and Fig. 20 for two of the 26 groups. The Ex. 3a data shown here was taken at axial level 8 where the axial flux profiles reach a maximum value. The fast flux peak in the fuel regions of the core is clearly visible in Group 1 (Fig. 19), whereas the peak thermal flux occurs in Group 20 and in the inner reflector as the fast neutrons are thermalized (Fig. 20).

As a final example, the average fuel temperature relative difference (%) between Ex. 2d and Ex. 3b is shown for the bottom (Fig. 21) and top (Fig. 22) axial layers of the core. In the hot bottom regions of the core the differences are less than -8% (i.e. the temperatures are higher for Ex. 2d than Ex. 3b), but these variations increase as the colder upper regions of the core are reached, where differences up to 23% are calculated for block 29 (Fig. 22).

A significant difference therefore exists between the axial temperature profiles of Ex. 2c and Ex. 3b (shown in Fig. 23 for block 9, as an example). This is caused by the feedback kinetics of the coupled case, as opposed to the constant power profile specified in Ex. 2d. The cross-section feedbacks on the xenon concentration, control rod position and the fuel and moderator temperatures all contribute to this change in the axial temperature distribution.

VI. CONCLUSIONS

Selected results for Phase I Exercises 2 and 3 of the OECD/NEA MHTGR-350 benchmark were presented for two models: a conventional homogenized “ring” model and a higher fidelity triangular-based “block” model. It was shown that the RELAP5-3D code can be applied to obtain detailed fuel and reflector temperature profiles in a prismatic HTGR core. Although a significant penalty was paid in terms of the computational resources required for the block model, the resultant gain in spatial fidelity justifies the development of such detailed models. Local temperature hot spots and variances between the blocks can be identified with the use of such a refined model, which can be of significant importance in reactor safety studies, where the margins to fuel temperature limits (for example) need to be quantified.

In the next phase of the benchmark, the two models will again be compared for two loss-of-cooling transients, two water ingress events and a power load follow transient. It is expected that the higher spatial fidelity of the block model will be especially apparent during the reactivity insertion caused by the water ingress into the MHTGR core.

ACKNOWLEDGMENTS

This work was supported by the U.S. Department of Energy, Assistant Secretary for the Office of Nuclear Energy, under DOE Idaho Operations Office Contract DE-AC07-05ID14517.

REFERENCES

1. J. ORTENSI, et al., "Prismatic Core Coupled Transient Benchmark", *Trans. of the American Nuclear Society*, **104**, p. 854 (2011).
2. J. ORTENSI, et al., "Prismatic Coupled Neutronics/Thermal Fluids Transient Benchmark of the MHTGR-350 Core Design: Benchmark Definition", Draft version 07/16/2013, Organization for Economic Corporation and Development (2013).
3. G. STRYDOM, A. EPINEY, "Relap5-3D results for Phase I (Exercise 2) of the OECD/NEA MHTGR-350 MW benchmark", *Proc. of conf. ICAPP 2012*, Chicago, USA (2012).
4. G. STRYDOM, et al., "INL Results for Phases I and III of the OECD/NEA MHTGR-350 Benchmark", INL/EXT-13-30176, Idaho National Laboratory (2013).
5. A. EPINEY, et al., "New Multi-group Transport Neutronics (PHISICS) Capabilities for RELAP5-3D and its Application to Phase I of the OECD/NEA MHTGR-350 MW Benchmark", *Proc. of conf. HTR 2012*, Tokyo, Japan (2012).
6. C. RABITI, et al., "PHISICS: A New Reactor Physics Analysis Toolkit", Transactions of the American Nuclear Society, Vol. 104, Hollywood, USA (2011).
7. RELAP5-3D[®] Code Manual, Vol.1-5, Rev. 3, INEEL-EXT-98-00834, Idaho National Laboratory (2009).
8. "PBMR coupled neutronics/thermal hydraulics transient benchmark: the PBMR-400 core design. Volume I: Benchmark Definition". NEA/NSC/DOC(2013)10, Organization for Economic Corporation and Development (2013).
9. P. J. TURINSKY et al., "NESTLE: A Few-Group Neutron Diffusion Equation Solver Utilizing the Nodal Expansion Method for Eigenvalue, Adjoint, Fixed-Source Steady

State and Transient Problems”, EGG-NRE-11406, Idaho National Engineering Laboratory, (1994).

- 10.Y. WANG, et al., “Improved Neutronics Treatment of Burnable Poisons for the Prismatic HTR”, *Proc. of conf. HTR 2012*, Tokyo, Japan (2012).
- 11.Y. WANG, C. RABITI, G. PALMIOTTI, “Krylov Solvers Preconditioned with the Low-Order Red-Black Algorithm for the PN Hybrid FEM for the INSTANT Code,” *Proc. of conf. M&C 2011*, Rio de Janeiro, Brazil (2011).
- 12.A. ALFONSI, et al., “PHISICS Toolkit: Multi-Reactor Transmutation Analysis Utility–MRTAU”, *Proc. of conf. PHYSOR 2012*, Knoxville, Tennessee, USA (2012).
- 13.A. EPINEY, et al., “PHISICS multi-group transport neutronic capabilities for RELAP5”, *Proc. of conf. ICAPP 2012*, Chicago, USA (2012).
- 14.C. RABITI, A. ALFONSI, A.S. EPINEY, “New Simulation Schemes and Capabilities for the PHISICS/RELAP5-3D Coupled Suite”, *in publication*, Nuclear Science and Technology (2015).
- 15.G. STRYDOM, “TINTE uncertainty analysis of the maximum fuel temperature during a DLOFC event for the 400 MW Pebble Bed Modular Reactor”, *Proc. of conf. ICAPP 2004*, Pittsburgh, USA (2004).

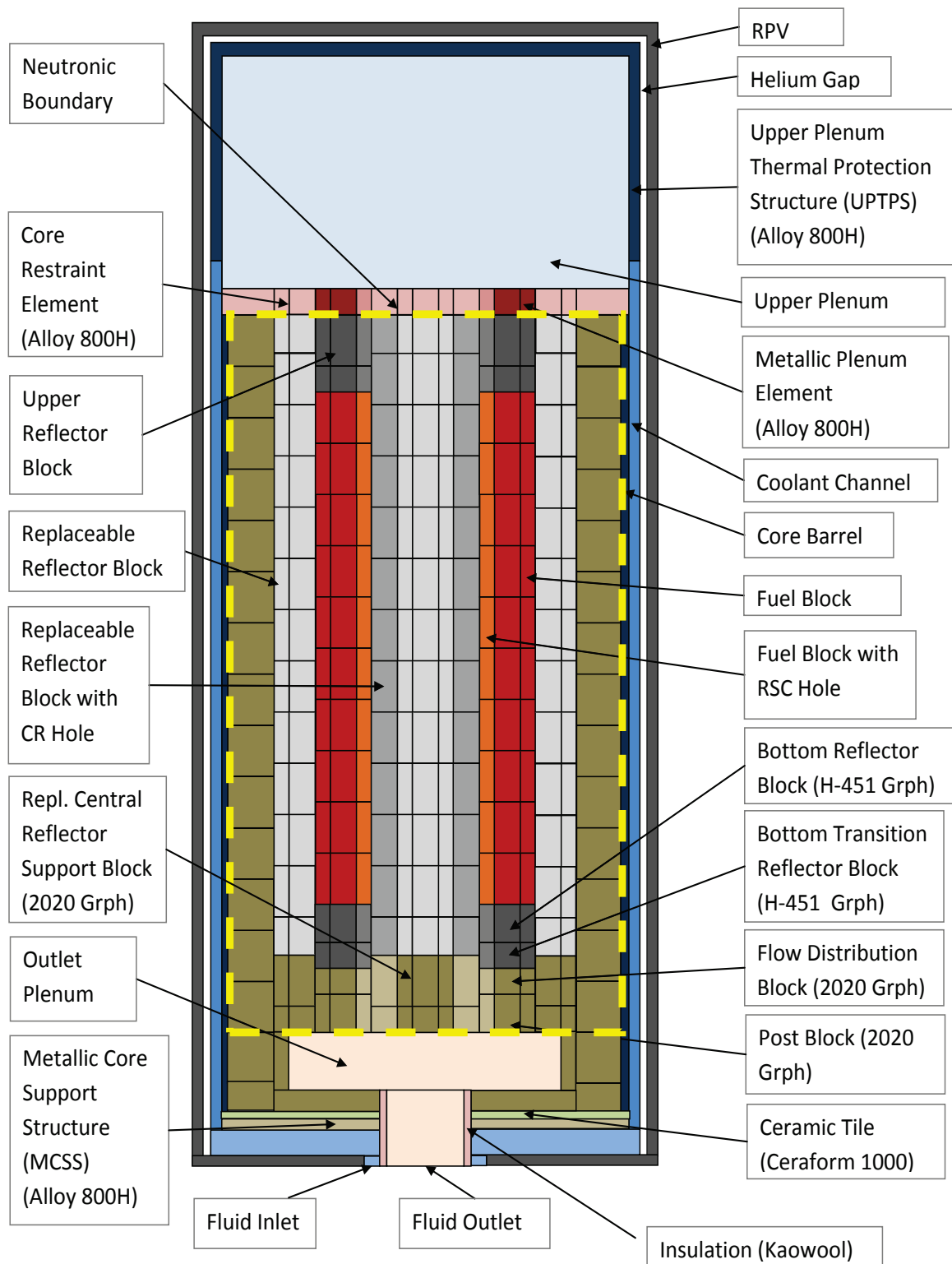


Fig. 1. MHTGR-350 Core Axial Layout.

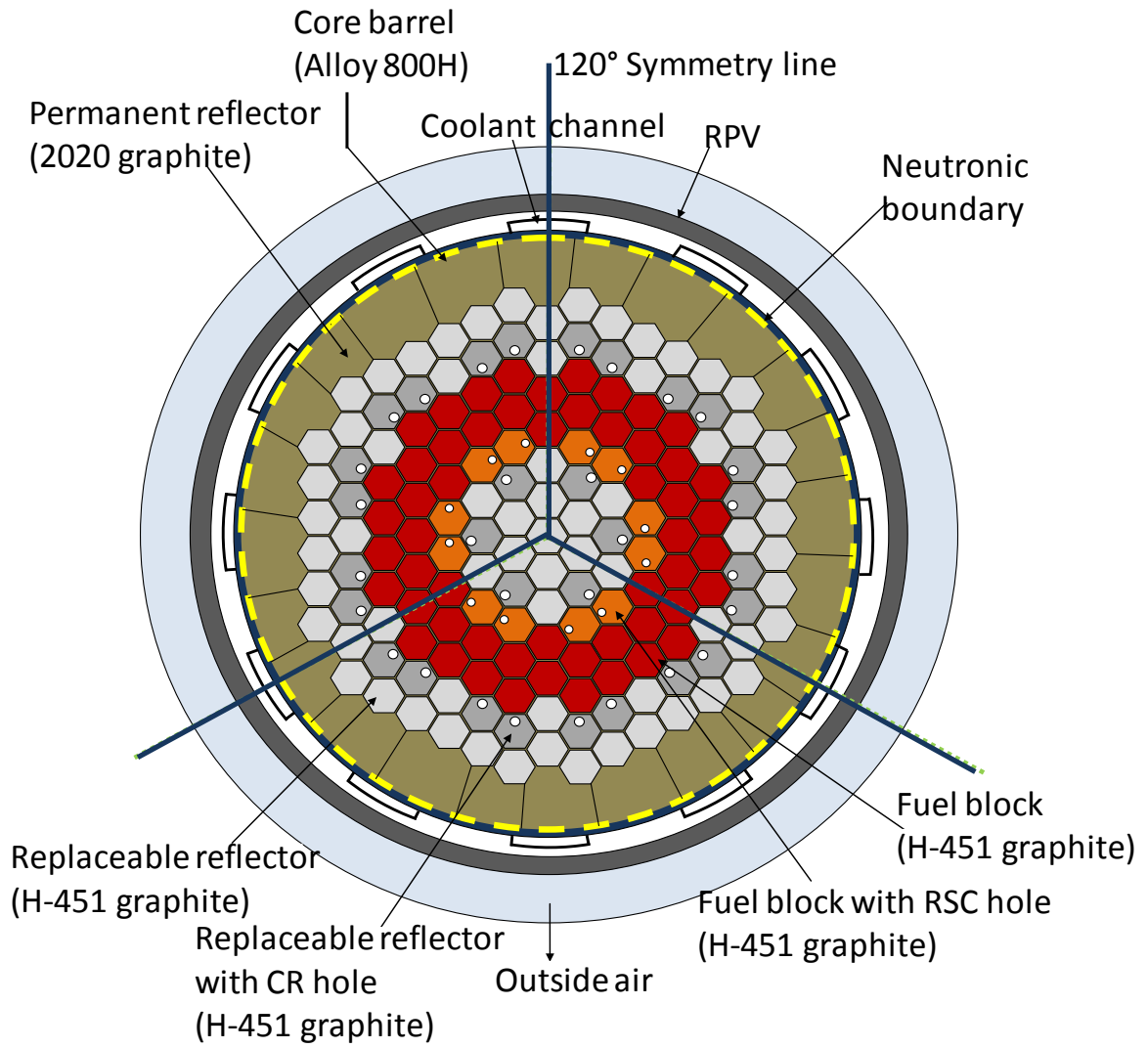


Fig. 2. MHTGR-350 Core Radial Layout.

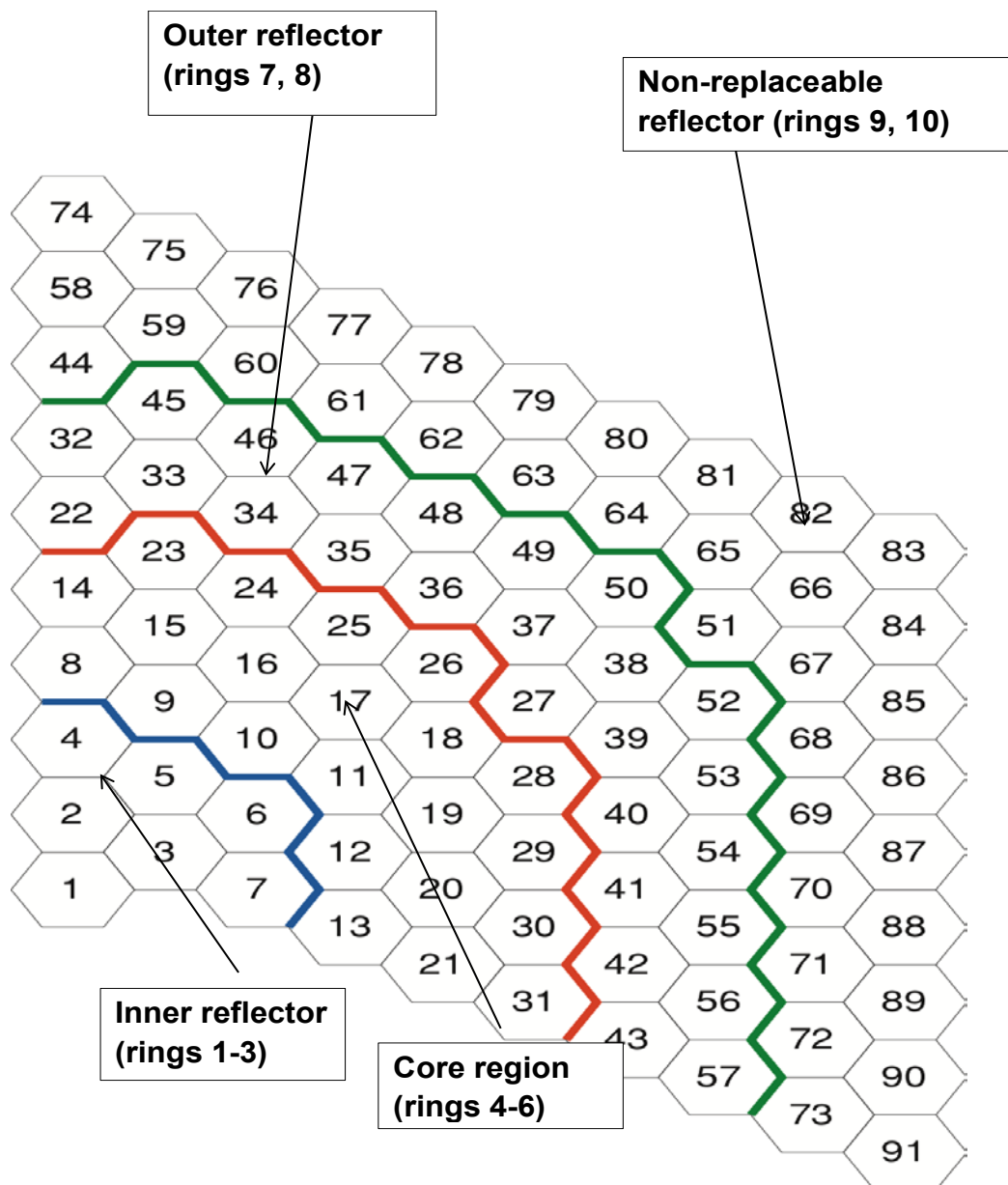


Fig. 3. Core Numbering Layout (axial layer 1: bottom of core).

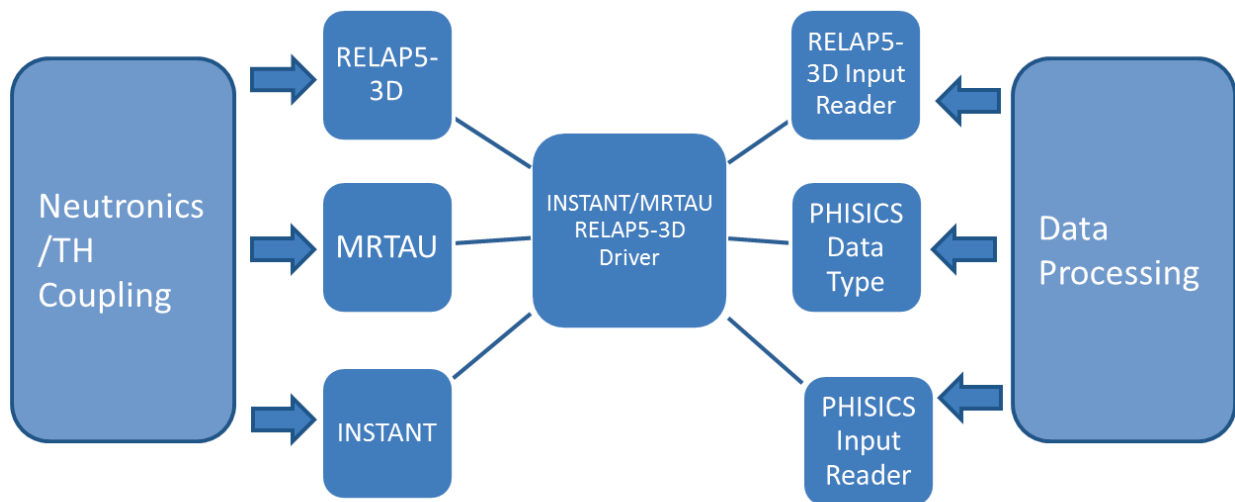


Fig. 4. PHISICS/RELAP5-3D Coupling Structure.

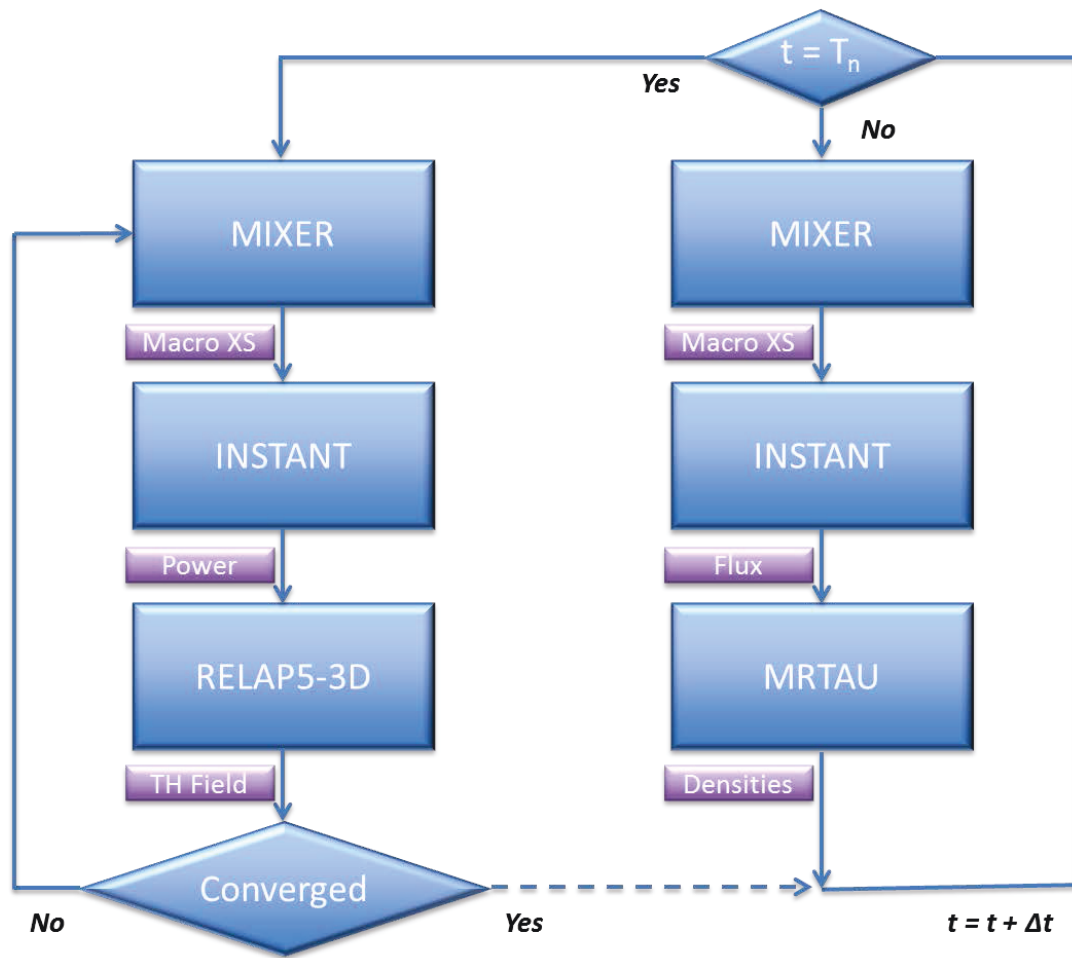


Fig. 5a. Depletion Time Evolution Scheme.

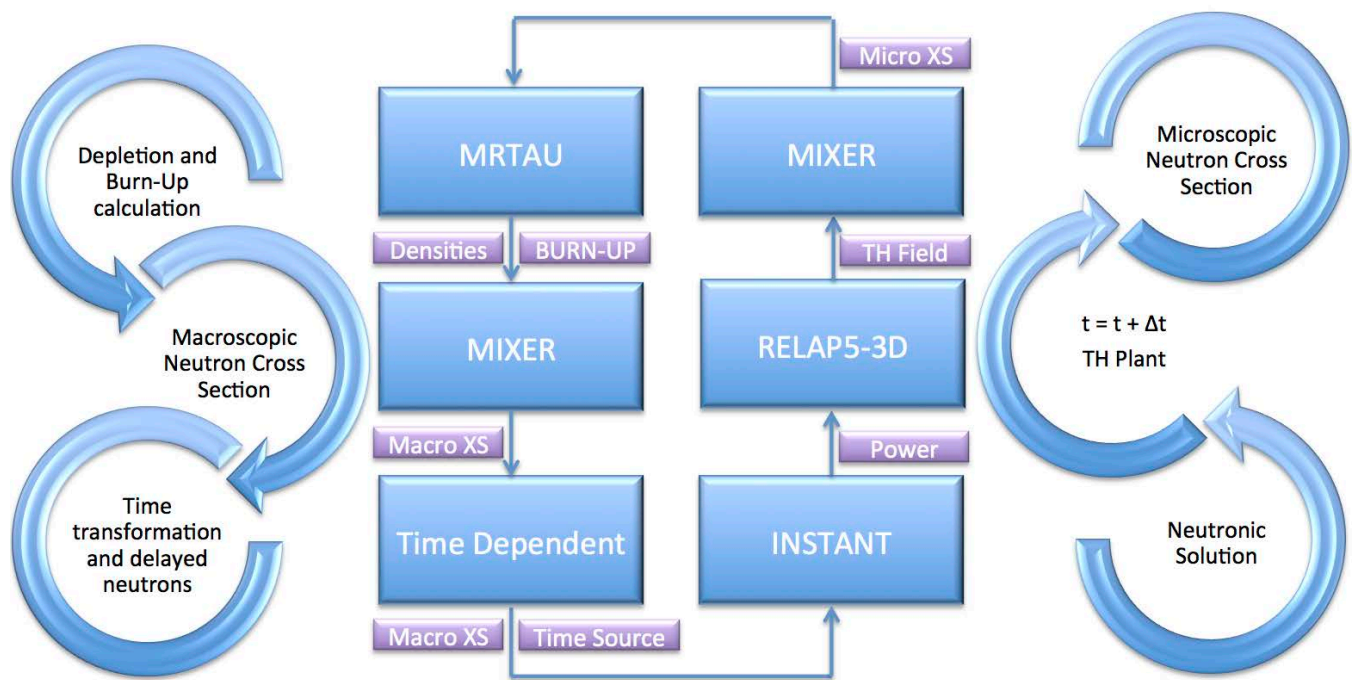


Fig. 5b. Time-Dependent Scheme.

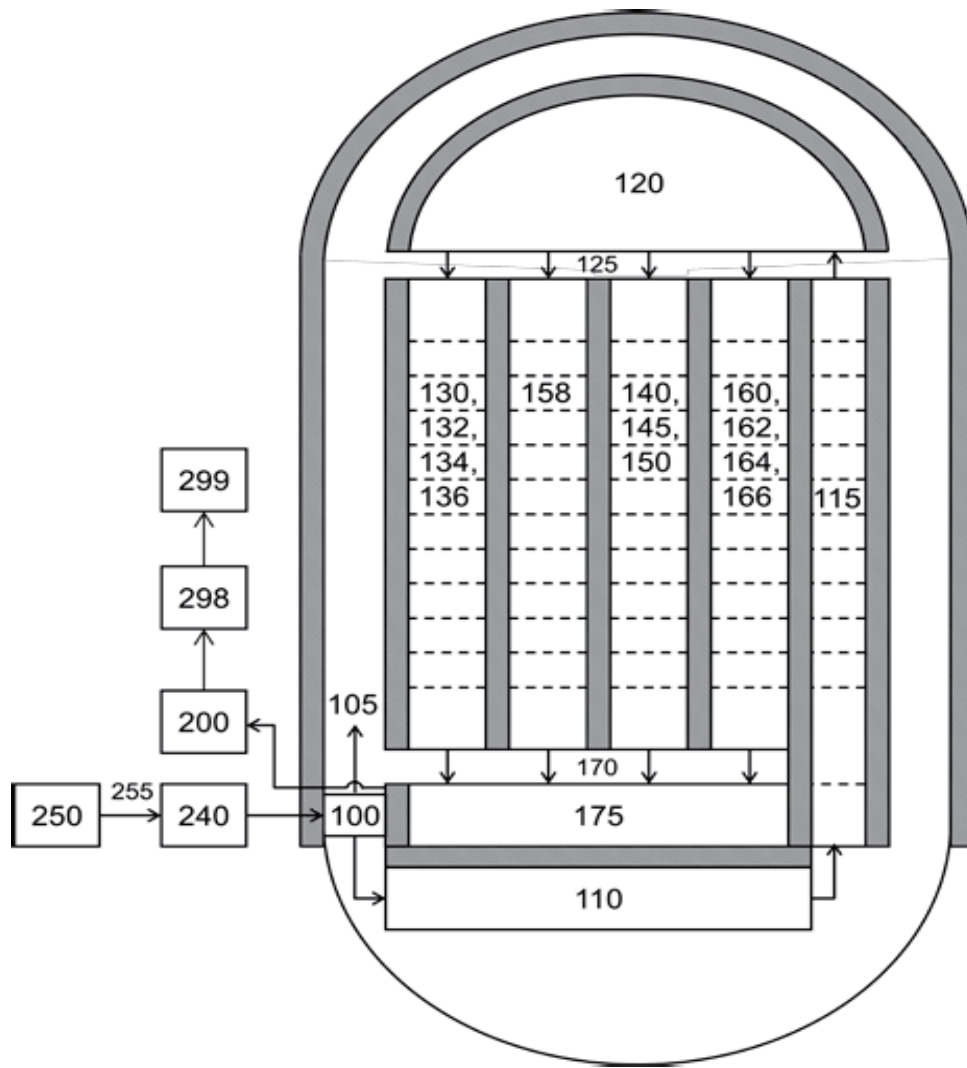


Fig. 6. RELAP5-3D Reactor Vessel Model Nodalization.

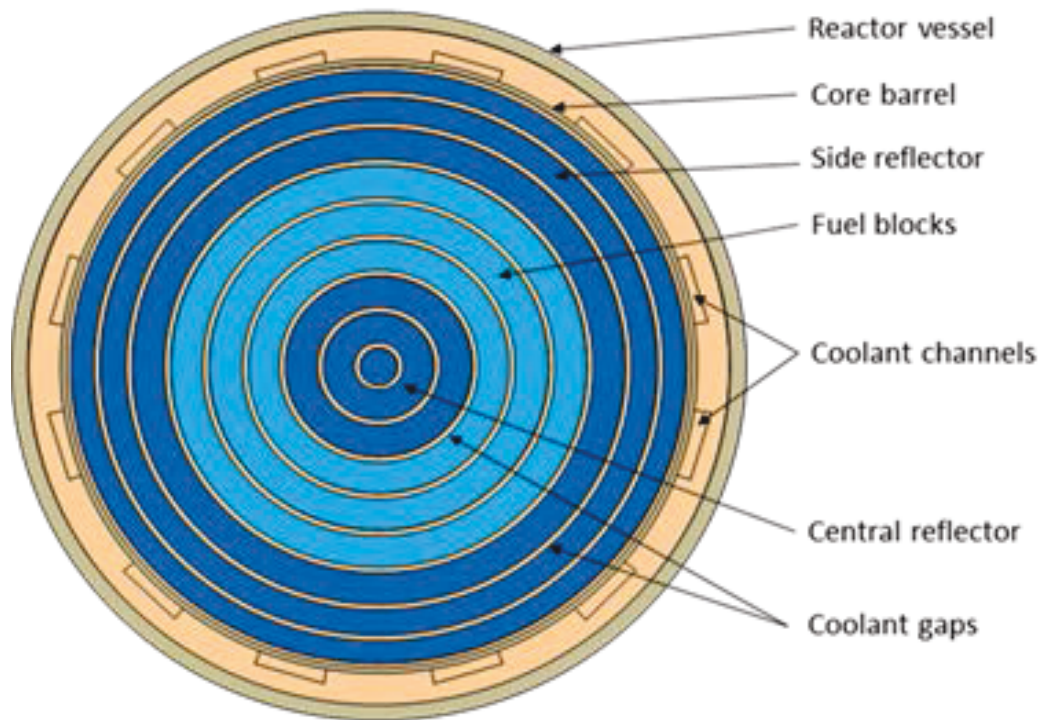


Fig. 7: Ring Model Radial Representation.

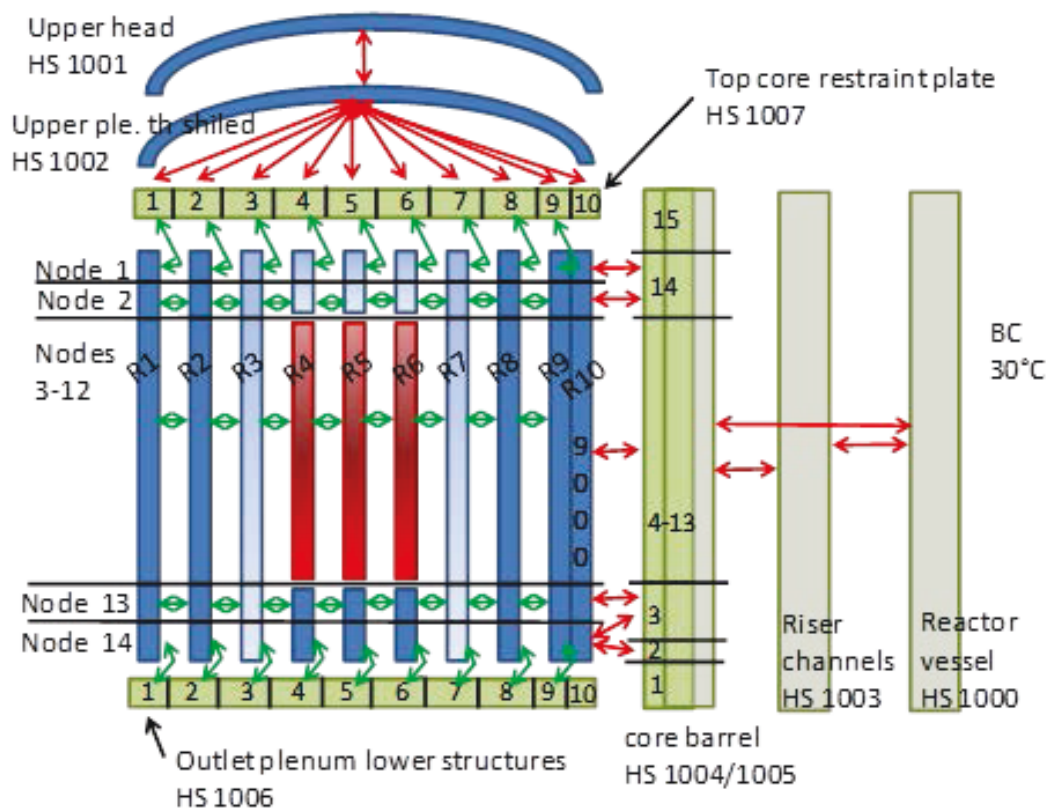


Fig. 8. Conduction (green arrows) and Radiation (red arrows) Enclosures in the RELAP5-3D Model.

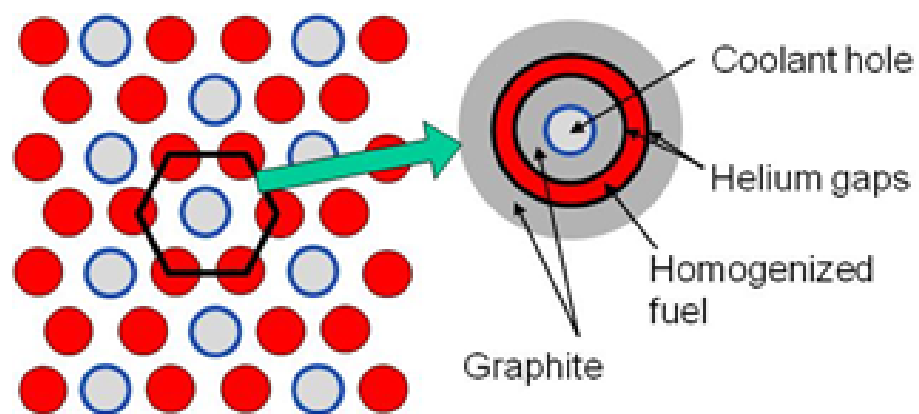


Fig. 9. Unit-cell Configuration.

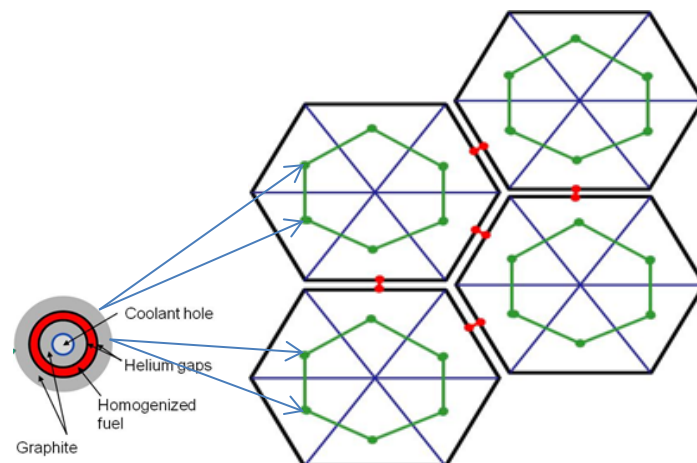


Fig. 10. Block Model Conduction and Radiation Connections.

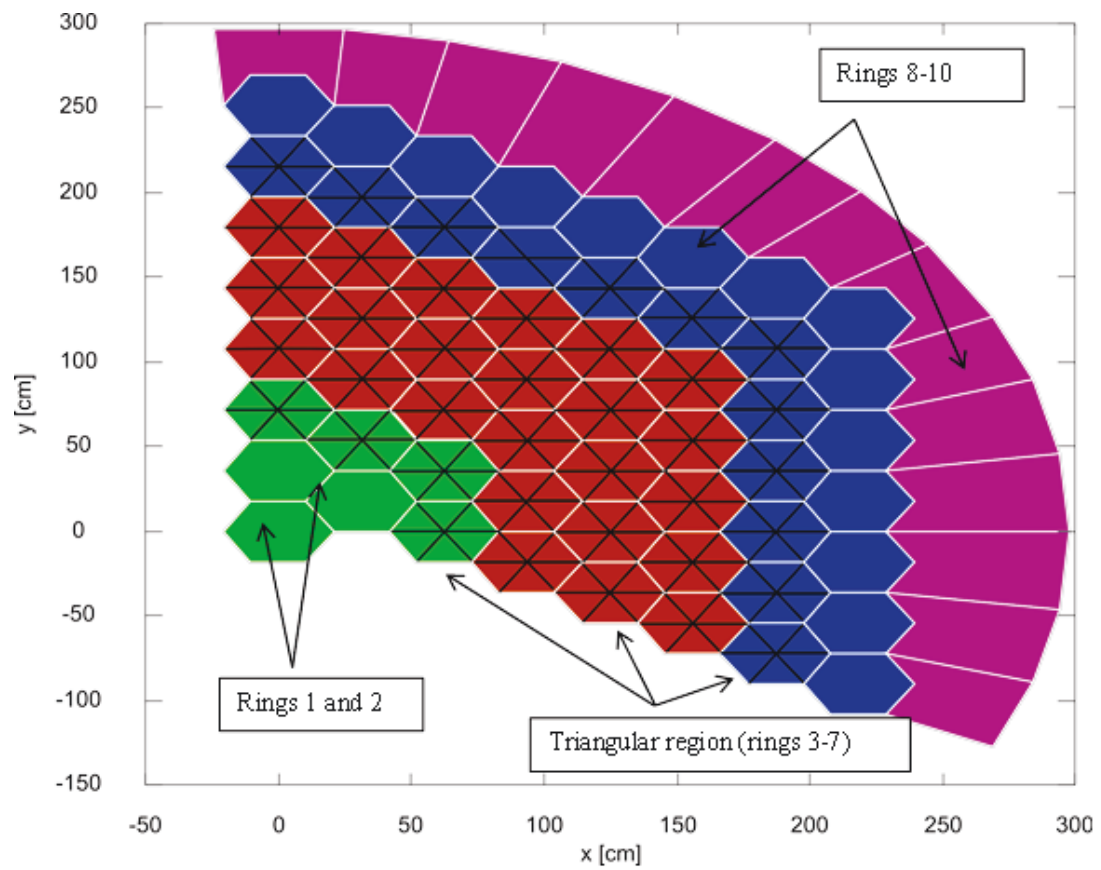


Fig. 11. Ring and Triangular Geometry Utilized in the RELAP5-3D Block Model.

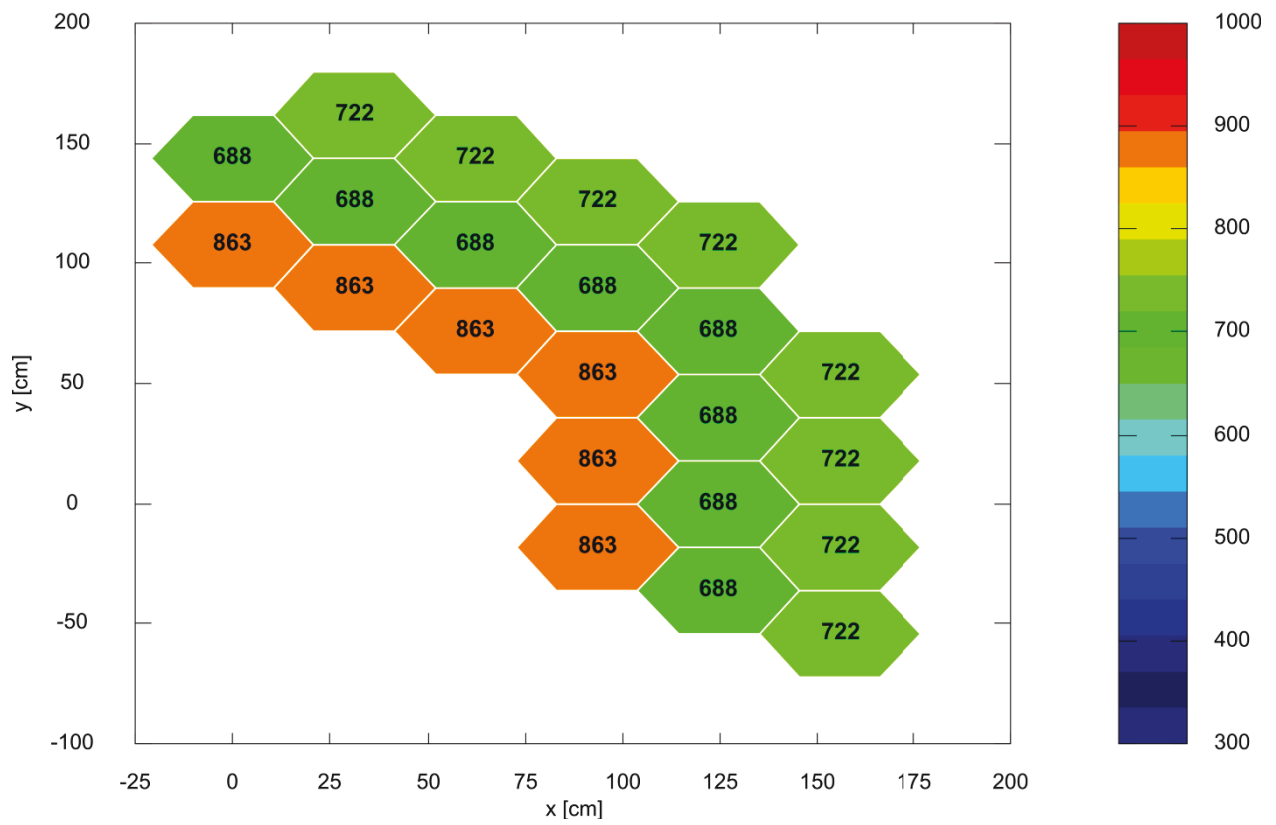


Fig. 12. Ex. 2a – Average Fuel Temperature (°C) Distribution on Axial Level 1 (Bottom Core) for the Ring Model.

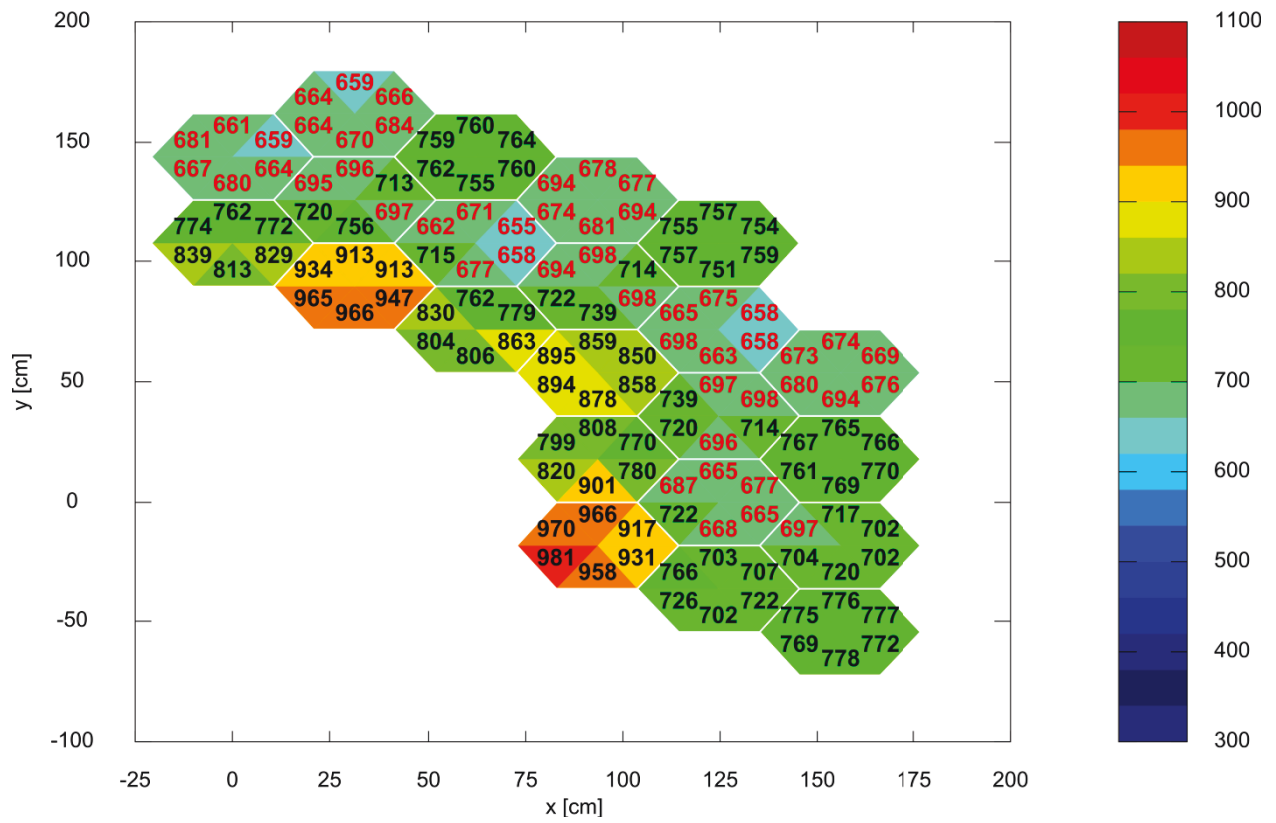


Fig. 13. Ex. 2a – Average Fuel Temperature (°C) Distribution on Axial Level 1 for the Block Model.

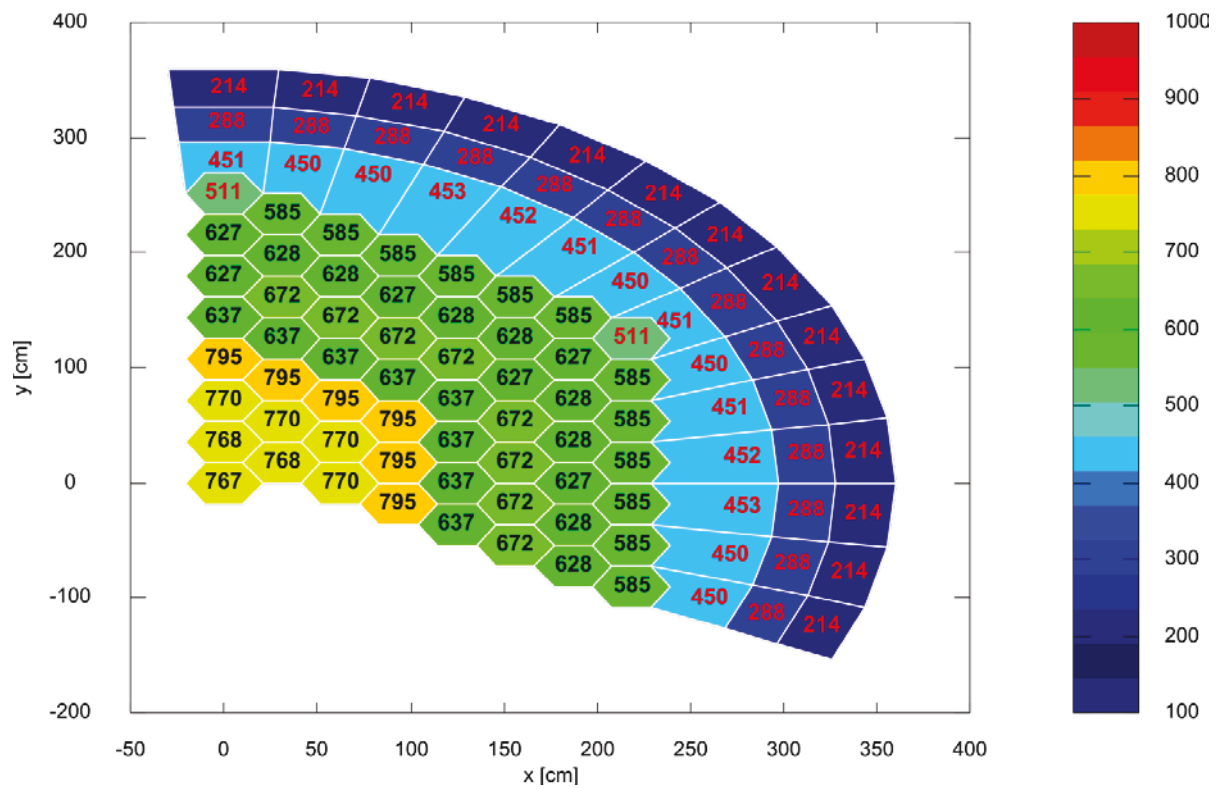


Fig.14. Ex. 2a – Average Graphite Temperature (°C) Distribution for all Graphite Structures in the Ring Model on Axial Level 1.

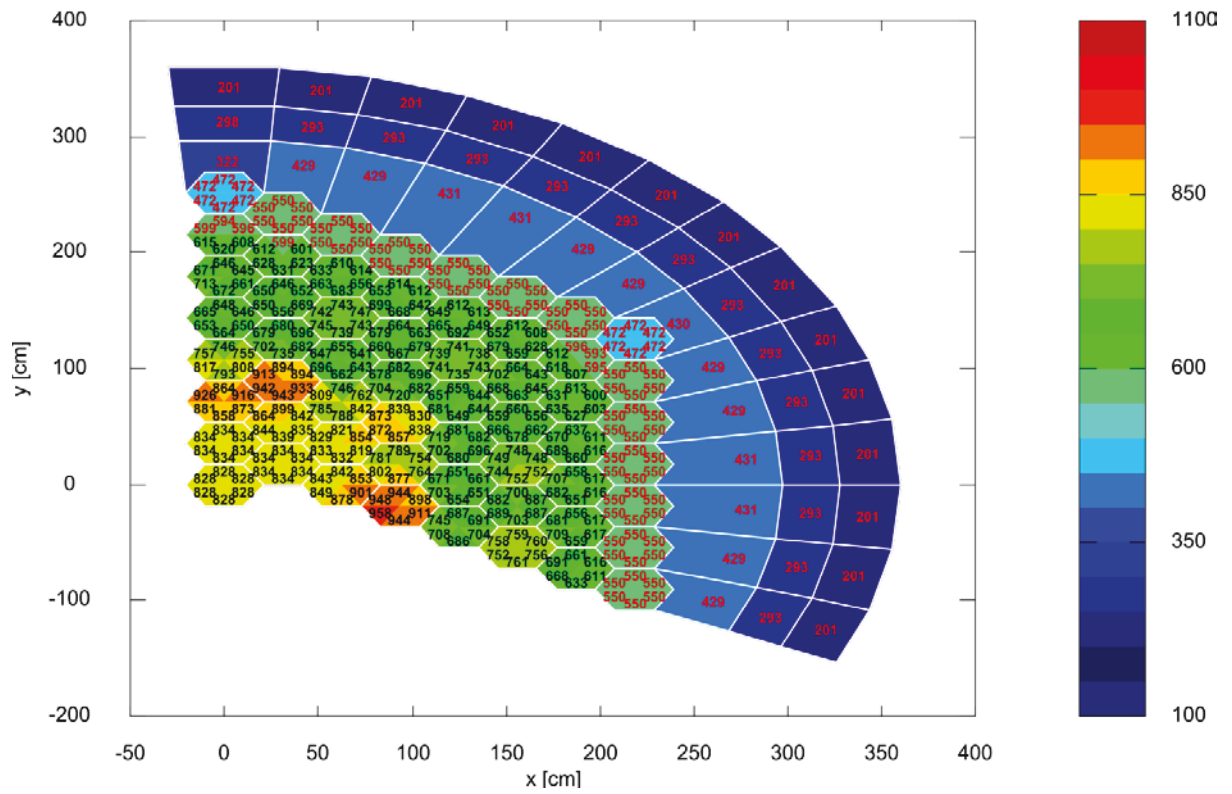


Fig. 15. Ex. 2a – Average Graphite Temperature (°C) Distribution for all Graphite Structures in the Block Model on Axial Level 1.

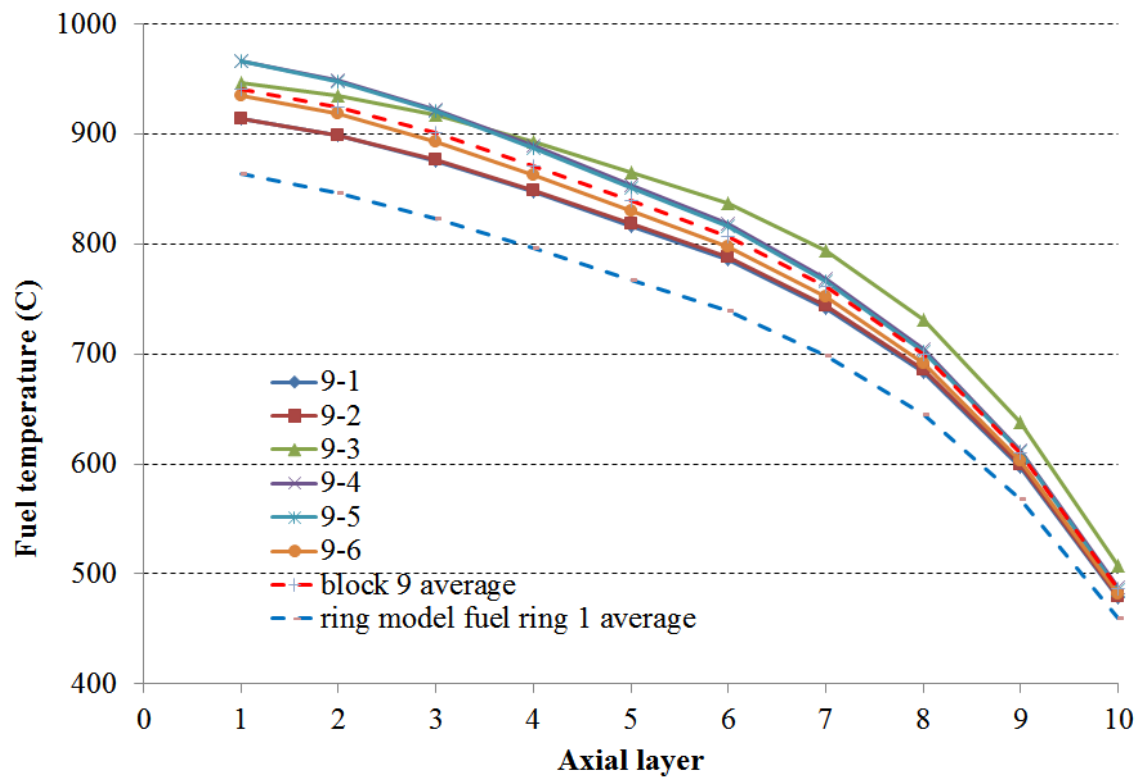


Fig.16. Ex. 2a – Block 9 Triangular Element Average Fuel Temperatures (°C).

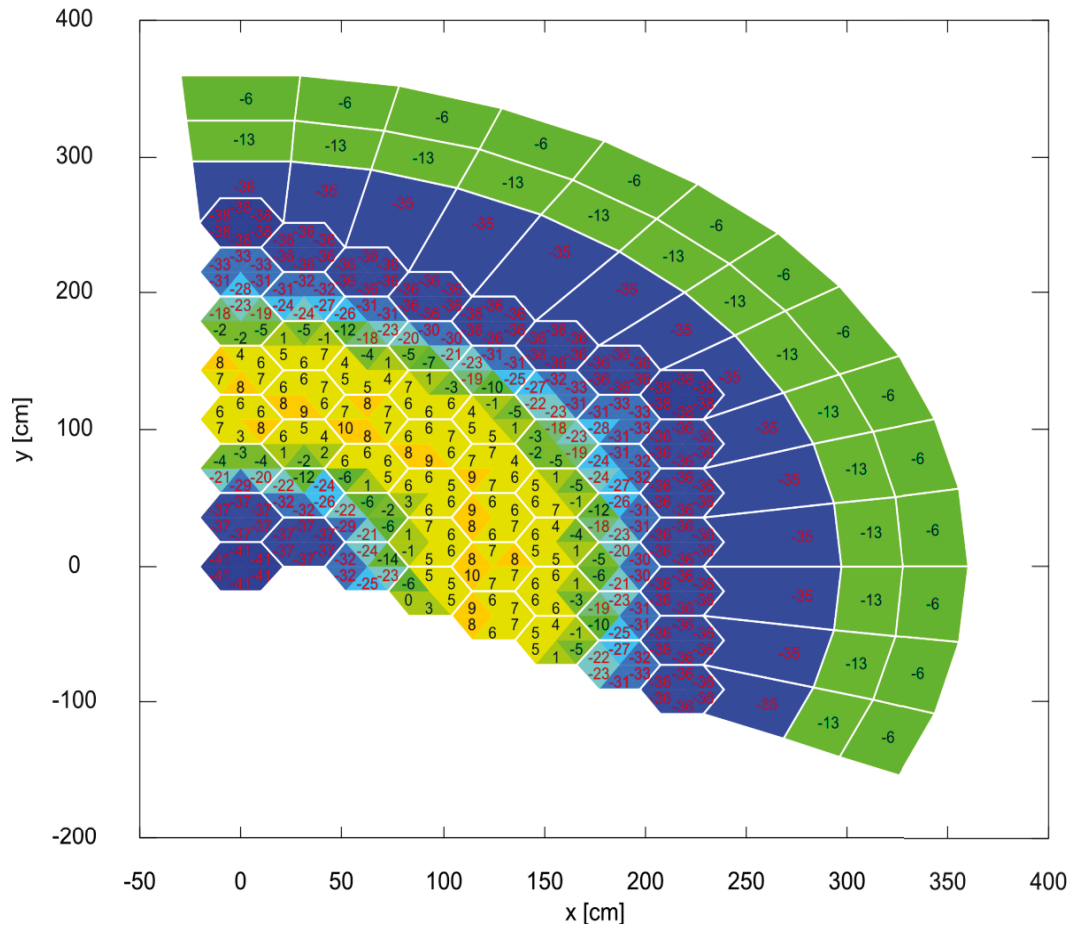


Fig.17. Average Graphite Temperature Difference (%) between Ex. 2a And Ex. 2b for all Graphite Structures in the Block Model on Axial Level 1.

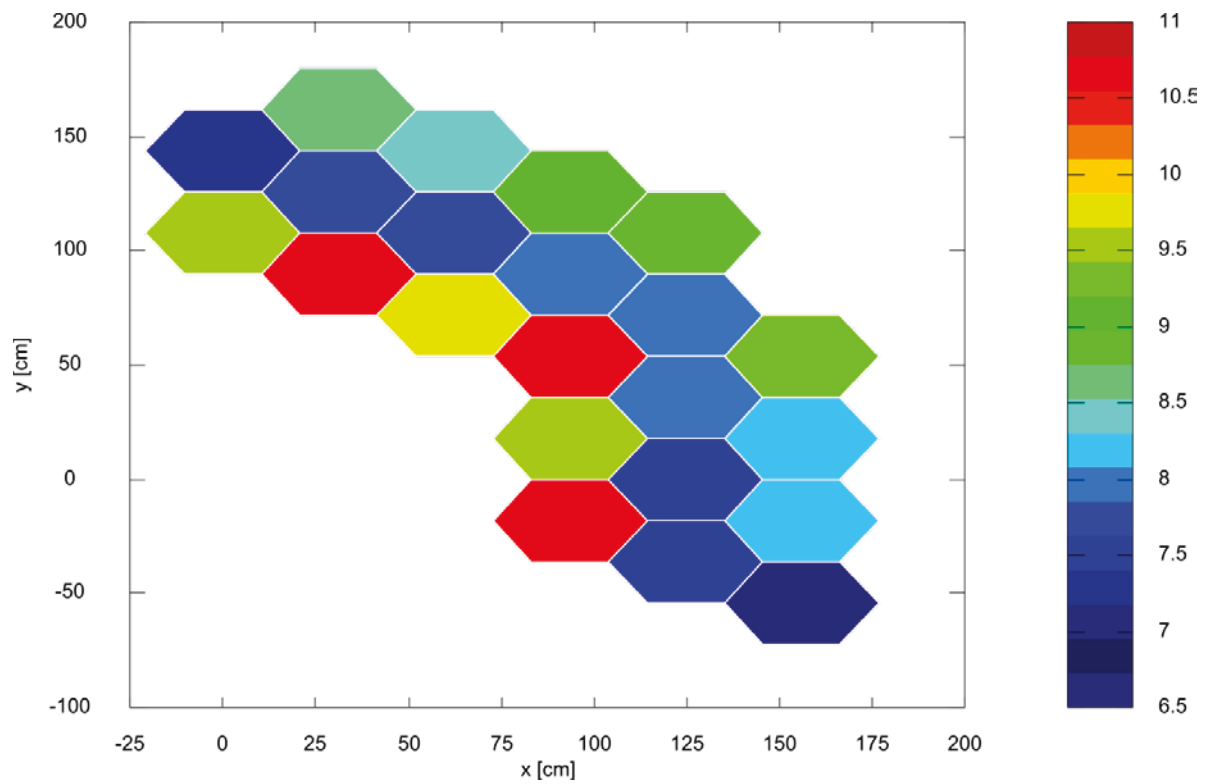


Fig. 18. Ex. 3a - Power Density (W/Cm^3) Distribution on Axial Level 8 for the Block Model.

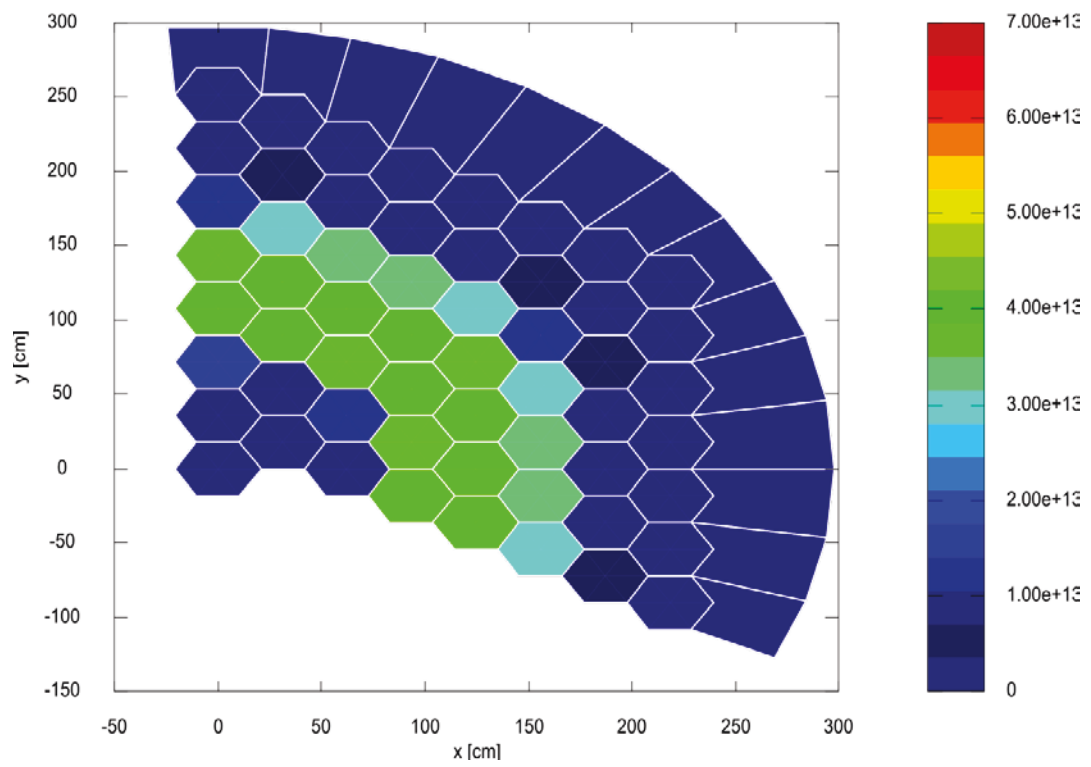


Fig. 19. Ex. 3a – Group 1 Flux Distribution on Axial Level 8 for the Block Model (Fast Peak Value).

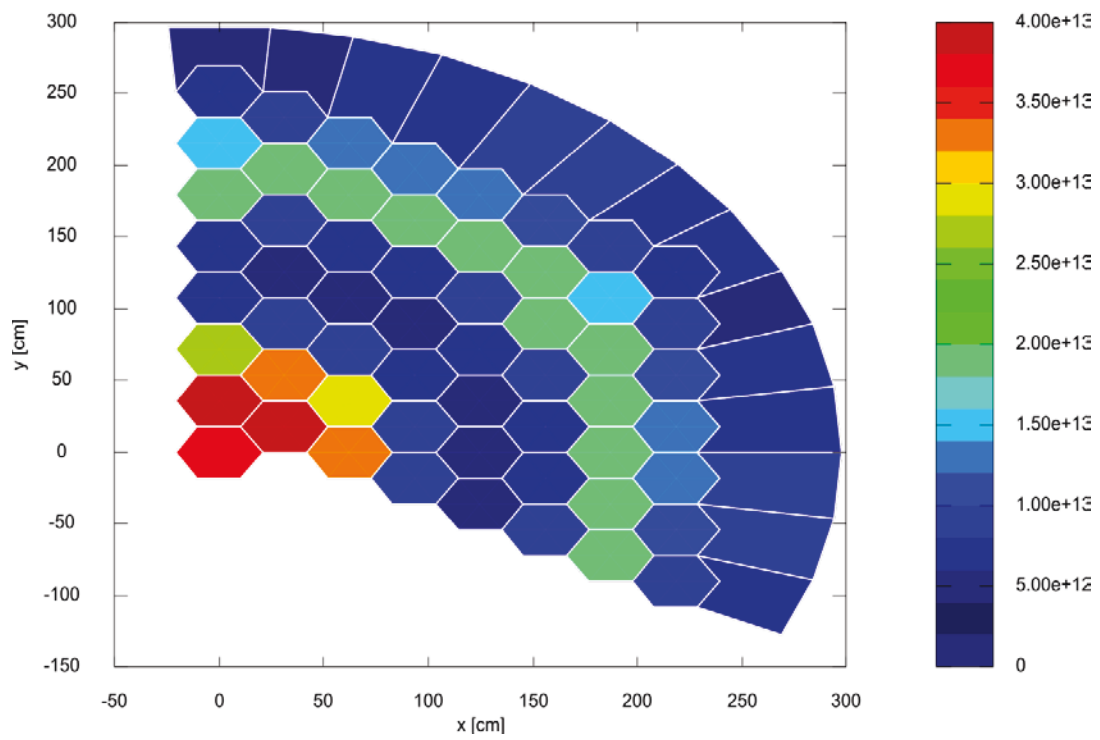


Fig. 20. Ex. 3a – Group 20 Flux Distribution on Axial Level 8 for the Block Model
(Thermal Peak Value).

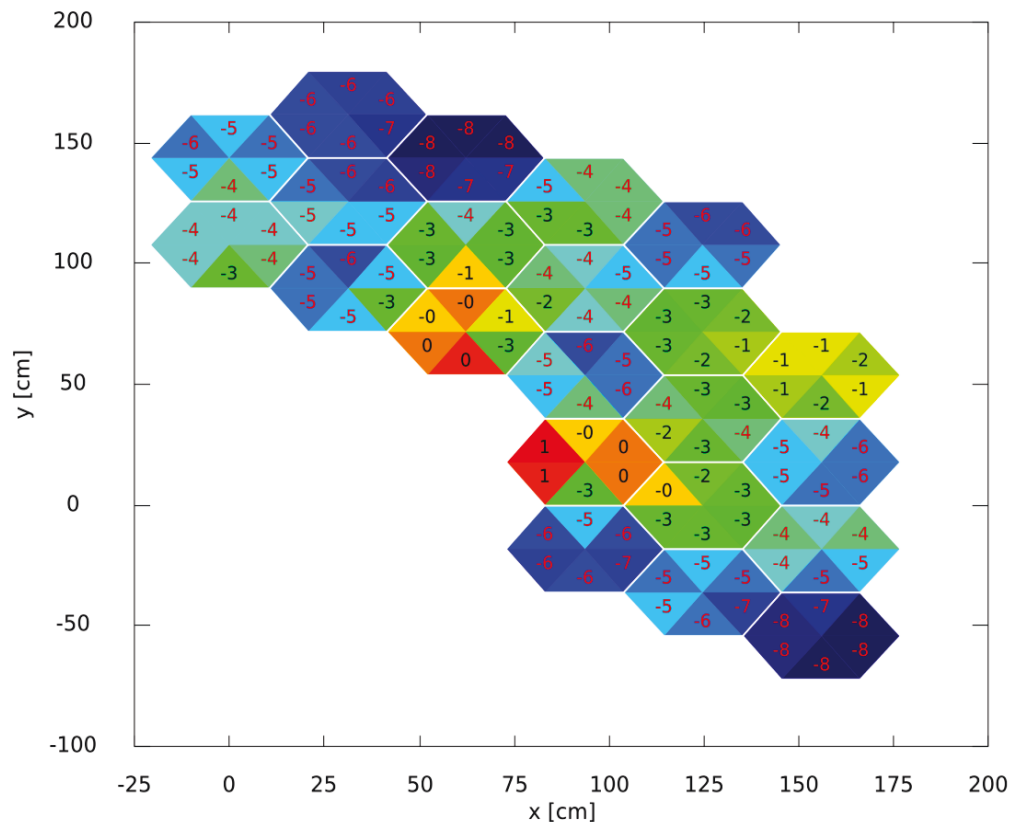
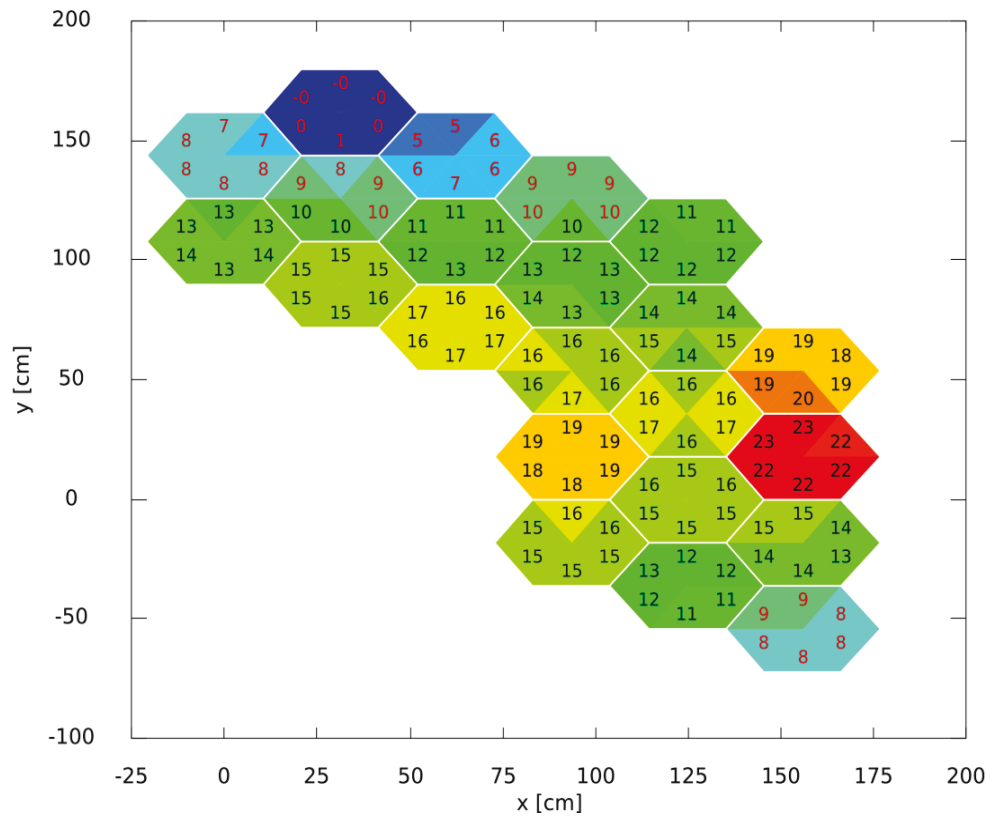


Fig. 21. Average Fuel Temperature Difference (%) between Ex. 2d and Ex. 3b on Axial Level 1 (First Bottom Core Level) for the Block Model.



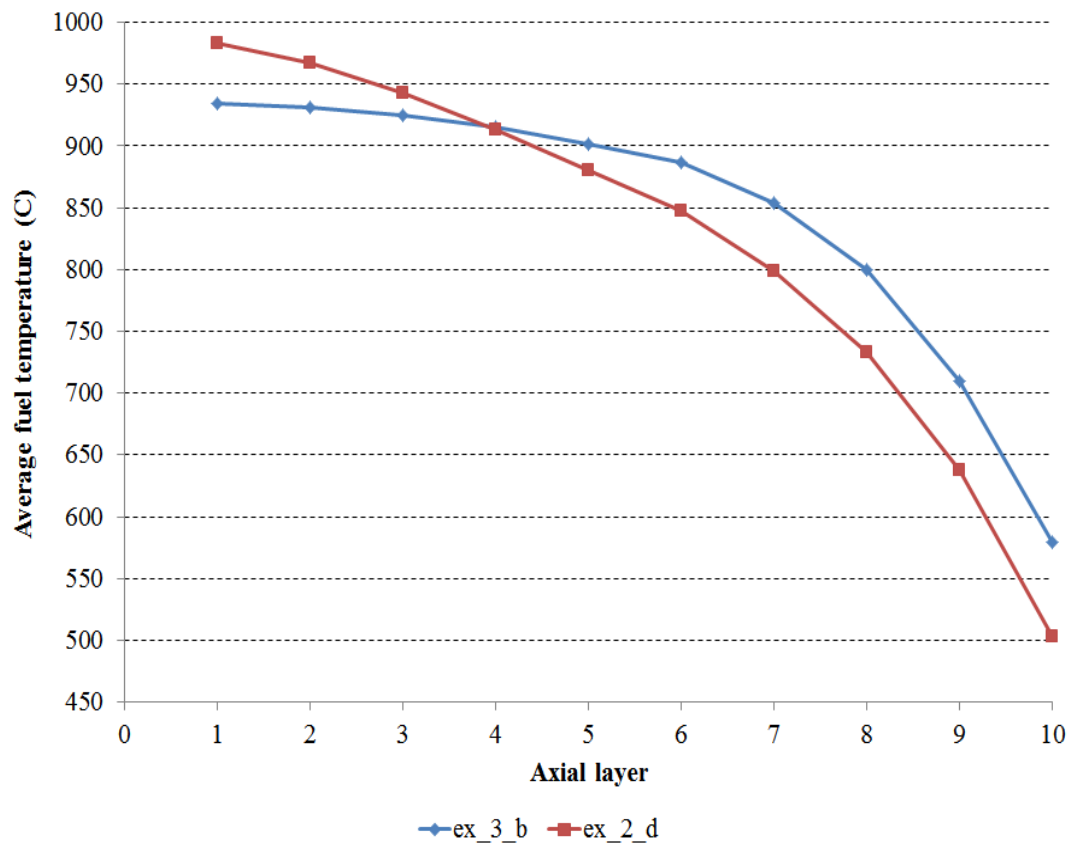


Fig. 23. Comparison of Block 9 Axial Average Fuel Temperature Profiles for Ex. 2d and Ex. 3b.

Table I. MHTGR-350 Main Characteristics.

Description	Value
Power	350 MW(thermal)
Core/fuel design	Graphite moderated, 660 prismatic hex-blocks with 15.5 wt% enriched UCO TRISO fuel compacts.
Coolant	Helium @ 6.39 MPa
Core inlet/outlet gas temperature	259°C / 687°C

TABLE II. Comparison of Ring and Block Model Metrics.

Description	Ring model	Block model
Hydrodynamic components	~20	~250
Heat structures	~20	~1,100
Materials	9	~100
Kinetic feedback zones	170	~4,700
Radiation/conduction enclosures	17 enclosures with ~350 surfaces	88 enclosures with ~7,000 surfaces
Decay heat treatment	1 table (global core power)	220 tables (block power)
Input file size (number of lines)	~30,000	~95,000
Real time required for converged steady state (hours)	<i>Ex. 2:</i> 0.1 <i>Ex. 3:</i> 11	<i>Ex. 2:</i> 5 <i>Ex. 3:</i> 20

TABLE III. Ex. 2a-2d Definitions

Exercise	Bypass flow	Thermophysical properties
2a	None	Fixed
2b	Fixed at 11% of total, distributed to 7 channels	Fixed
2c	Fixed at 11% of total, distributed to 7 channels	Variable
2d	Explicit, based on 2 mm gaps between blocks	Variable

Table VI: Material properties for Ex. 2a and Ex. 2b

Material	Conductivity (W/cm.K)	Emissivity
Fuel compact	20.0	0.85
Fuel block	37.0	0.85
Grade H-451 graphite	66.0	0.85
Grade 2020 graphite	35.0	0.85
Core barrel		
Core restraint element	17.8	0.85
Upper plenum thermal protection system		
Ceramic tile	0.91	0.85
Thermal insulation	0.175	0.85
RPV	40.0	0.85
Coolant	0.41	-
Air	0.068	-
Heat sink	-	0.8

Table V: Bypass Flow Distribution for Ex. 2b and Ex. 2c

	Component	%
1	In-core	1.50
2	Inner Reflector	0.50
3	Inner Control Rod Cooling	1.20
4	Outer Control Rod Cooling	1.80
5	Outer Reflector (First Ring)	1.38
6	Outer Reflector (Second Ring)	1.62
7	Permanent Side Reflector	3.00
	Total	11.00

TABLE VI. Ex. 3a and Ex. 3b Comparison of PHISICS/RELAP5-3D k_{eff} and CR Worths for the Ring and Block models.

PHISICS- RELAP5-3D model	CR nominal	CR all out	CR all in	CR worth delta k/k (pcm)
<i>3a</i>				
ring	1.03447	1.03593	1.02673	888
block	1.03394	1.03511	1.02513	965
<i>delta k (pcm)</i>	<i>53</i>	<i>81</i>	<i>160</i>	
<i>3b</i>				
ring	1.03336	1.03450	1.02493	926
block	1.03384	1.03503	1.02549	922
<i>delta k (pcm)</i>	<i>-48</i>	<i>-52</i>	<i>-56</i>	